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The EU's External Raw Materials Strategy: Key Fields for Action

Building Resilient Partnerships and Advancing Local Value Creation

Inga Carry, Melanie Müller, and Meike Schulze

With the participation of Veronika Jall, Lea Strack, and Mark Schrolle

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Abbreviations

AfCFTA	African Continental Free Trade Area
AfDB	African Development Bank
AMDC	African Minerals Development Centre
AMV	Africa Mining Vision
AU	African Union
BRI	Belt and Road Initiative
BRICS	Brazil, Russia, India, China, South Africa
CBAM	Carbon Border Adjustment Mechanism
CEPAL	Comisión Económica para América Latina y el Caribe
CRM	Critical Raw Material
CRM Board	European Critical Raw Materials Board
CRMA	Critical Raw Materials Act
DGs	Directorates-General
DG GROW	Directorate-General for Internal Market, Industry, Entrepreneurship and SMEs
DG INTPA	Directorate-General for International Partnerships
DG TRADE	Directorate-General for Trade
DRC	Democratic Republic of the Congo
EBRD	European Bank for Reconstruction and Development
ECLAC	Economic Commission for Latin America and the Caribbean
ECOWAS	Economic Community of West African States
EEAS	European External Action Service
EFSD+	European Fund for Sustainable Development Plus
EIB	European Investment Bank
ESG	Environmental, Social, Governance
EU	European Union
EV	Electric Vehicle
FDI	Foreign Direct Investment
FTA	Free Trade Agreement
GCC	Gulf Cooperation Council
GIZ	Deutsche Gesellschaft für Internationale Zusammenarbeit
G7	Group of Seven
G20	Group of Twenty
IEA	International Energy Agency
IGF	Intergovernmental Forum on Mining, Minerals, Metals and Sustainable Development
IRA	US Inflation Reduction Act
JETP	Just Energy Transition Partnership
LCP	Local Content Policy
LIB	Lithium-ion Battery
MERCOSUR	Southern Common Market
MoU	Memorandum of Understanding
MSP	Minerals Security Partnership
OECD	Organisation for Economic Co-operation and Development
REE	Rare Earth Element
RISE	Resilient and Inclusive Supply-chain Enhancement
R&D	Research and Development

SADC	Southern African Development Community
SWP	Stiftung Wissenschaft und Politik
TEI	Team Europe Initiative
UAE	United Arab Emirates
UK	United Kingdom
UNECA	United Nations Economic Commission for Africa
US	United States
WTO	World Trade Organization

Introduction

The political will within the European Union (EU) to strengthen economic security and diversify supply chains in strategic sectors has further solidified since the Russian invasion of Ukraine in 2022, and it is being reinforced by the intensifying geopolitical rivalry between the United States (US) and China. In this context, critical raw materials (CRMs) have moved to the forefront of the EU's strategic agenda. These materials are essential not only for maintaining the EU's global competitiveness in green, digital, and defence-related technologies and products, but also for mitigating risks associated with political dependencies and external coercion—risks that are particularly pronounced in sectors marked by a high reliance on imports such as critical minerals.

Since 2023, the level of intra-European coordination in the minerals sector has noticeably increased. Following a year-long legislative process, the Critical Raw Materials Act (CRMA) entered into force in May 2024, providing the legal foundation for a more active and coherent European engagement in the raw materials sector. The CRMA sets out clear benchmarks to expand mining, processing, and recycling in the EU, while also promoting the diversification of external supply sources. To support these objectives, the EU aims to establish strategic partnerships with resource-rich third countries. A central focus of the CRMA is the reduction of strategic dependencies—most notably on China, which currently accounts for approximately 60% of global production and 91% of the refined output for rare earth elements (REEs).[1] China also dominates the processing of other critical materials, refining around 70% of lithium, 78% of cobalt, and 45% of copper. The CRMA thus marks a clear shift from fragmented national efforts towards a unified European approach.

Under the new European Commission inaugurated in September 2024 under Ursula von der Leyen, the raw materials sector remains a priority within the broader framework of economic security. Moving forward, it will be crucial not only to ensure internal coherence within the Commission, but also to maintain the commitments of member states to a unified European approach—also in its external engagement.

Given the shifting global landscape in 2025, maintaining this collective momentum will be key to safeguarding Europe's strategic interests. The inauguration of President Donald Trump in January 2025 has heightened geopolitical uncertainty and placed additional strategic pressure on the EU. Escalating US-China tensions has already created disruptions in global mineral supply chains. In response to tariffs on Chinese imports, China imposed export restrictions on seven REEs and magnets. This has also affected the EU, with some industries facing shortages and potential production stops.[2]

[1] IEA, *Global Critical Minerals Outlook 2025*, Revised version (June 2025), https://iea.blob.core.windows.net/assets/ef5e9b70-3374-4caa-ba9d-19c72253bfc4/GlobalCriticalMineralsOutlook2025.pdf (accessed 23 June 2025).

[2] Gracelin Baskaran and Meredith Schwartz, *The Consequences of China's New Rare Earths Export Restrictions,* Critical Questions (Center for Strategic & International Studies, 14 April 2025), https://www.csis.org/analysis/consequences-chinas-new-rare-earths-export-restrictions (accessed 23 June 2025); Malte Kirchner, "China Trade Conflict: Rare Earths Are Becoming Rare – Now Also in the EU", *heise online*, 28 May 2025, https://www.heise.de/en/news/China-trade-conflict-Rare-earths-are-becoming-rare-now-also-in-the-EU-10418919.html (accessed 23 June 2025).

Beyond these immediate disruptions, the broader shift in US foreign policy presents a structural challenge for the EU. The Trump administration is widely expected to further prioritise unilateral approaches—such as tariff-based leverage, industrial subsidies, and selective decoupling—over multilateral cooperation. This undermines one of the cornerstones of the EU's past diversification efforts, which relied in part on transatlantic alignment and initiatives such as the Minerals Security Partnership (MSP). As the US becomes a less predictable actor, Europe will have to recalibrate its external strategy accordingly. At the same time, other actors—particularly the Gulf States—are actively strengthening their role in the global competition for access to critical minerals. These developments further reinforce the urgency for the EU to enhance its own strategic autonomy and resilience.

Yet, a significant share of current capacity is absorbed by short-term risk management, while geopolitical conflicts continue to amplify the EU's focus on resilience through the expansion of domestic capabilities. However, this focus alone will not be sufficient. Cooperation with, and imports from, third countries will remain critically important in the medium and long terms. In this context, the EU must not lose sight of its long-term objectives and should further strengthen the external dimension of its raw materials policy. A key challenge for the EU will be to maintain a strong partner orientation.

Partnerships with resource-rich countries in Africa, Latin America, and Asia will be particularly important. This working paper draws special attention to two case studies: Chile and Zambia—both mineral-rich countries from the so-called Global South, yet differing in their national and regional contexts, levels of economic development, and institutional capacities. Both countries have not only established strategic partnerships with the EU, but are actively redefining their raw materials strategies. In this, they are not alone.

In response to intensifying geopolitical competition, many mineral-rich countries are seeking to move beyond their traditional role as raw material exporters. Increasingly, they aim to retain a greater share of value creation by developing national industrial strategies and building local processing and manufacturing capacities. If the EU wants to position itself as a reliable partner, it must present attractive and credible offers—ones that are not only politically backed, but also supported by adequate financing and committed industry players that are ready to invest to enable sustainable outcomes. Only through meaningful engagement that aligns with the priorities of partner countries and promotes long-term local value creation can Europe realise the ambitious objectives set out in the CRMA. Yet, the external dimension of the CRMA remains underdeveloped. Therefore, this working paper explores how the EU can strengthen the international dimension of its raw materials policy, focusing on four interrelated areas of action.

Section 1 examines policies and concepts aimed at fostering local value creation. This issue has increasingly gained importance as many partner countries—such as Chile and Zambia—are developing national strategies to enhance domestic value addition and calling for stronger support from international partners. The EU has repeatedly committed to supporting such efforts in the context of its raw materials partnerships. In order for partnerships with developing and emerging economies to succeed, the EU must place the needs and priorities of its partners at the centre of its strategy. However, the political notion of "value creation" remains somewhat vague and underdefined. It requires a clearer conceptual framing as well as practical guidance for implementation. Promoting value creation in the raw materials sector involves more than expanding smelting or refining capacities. It must also raise critical questions about the technical and economic feasibility of developing downstream industries, especially given the wide range of

structural conditions across countries. The section highlights the important role of development cooperation in supporting partner countries in designing and implementing transformation strategies that are both economically viable and locally appropriate. At the same time, it emphasises that meaningful progress ultimately depends on strong engagement from the private sector.

Section 2 focuses specifically on strategies to promote local value creation in the critical minerals sector in Africa and Latin America. Both regions have demonstrated strong and proactive efforts to move up the value chain, and the EU has attributed strategic importance to both regions within its raw materials policy framework. As the section illustrates, effective cooperation requires more than unilateral offers. The EU must seriously engage with the interests and industrial policy strategies of its partners—some of which may conflict with its own strategic ambitions, such as the goal of relocating supply chains back to Europe. Alongside the growing number of national strategies to promote resource-based development, regional actors and initiatives are equally becoming more prominent in shaping global resource governance. For instance, the African Union (AU) has already responded to increasing global demand by placing greater emphasis on its own agendas for developing the raw materials sector. For large-scale, cross-border initiatives such as infrastructure corridors and economic integration to succeed, coordination at the regional level is essential, despite implementation challenges on the ground. For the EU, this underlines the importance of striking a balance between bilateral engagement and support for regional integration strategies. Without such balance, intensifying geopolitical competition risks fuelling rivalries among countries and reinforcing tendencies towards resource nationalism.

Section 3 examines the EU's efforts to date with regard to the external dimension of its raw materials strategy, focusing on its key instrument: the bilateral raw materials partnerships. It identifies several gaps that hinder the successful implementation of strategic cooperation with third countries. The section draws again primarily on the cases of Chile and Zambia. The foundations of these partnerships—underpinned by political commitments through Memoranda of Understanding (MoUs) and a gradually strengthening basis of technical cooperation—are in place and represent an important first step. But a critical shortcoming lies in the absence of sufficient industrial backing. Successful implementation requires much stronger engagement from the private sector. However, meaningful involvement from European companies has so far largely failed to materialise. The section underscores the need for more pressure by European policymakers on the private sector, alongside increased incentives and public investment to enable more effective implementation. Crucially, the partnerships' success hinges on improved cross-departmental coordination within the EU, as well as firm commitments from member states to support joint action in partner countries. Strengthening the "Team Europe" approach will be essential to enhancing the EU's credibility and impact as a raw materials partner.

Section 4 analyses the EU's external raw materials strategy in the context of an increasingly fragmented and politicised global landscape. As efforts to diversify supply chains accelerate, the EU finds itself navigating a complex web of actors, competing strategic interests, and intensifying systemic rivalry. One major disruption stems from the recent shifts in US priorities following the re-election of Trump. Although the Biden administration had championed cooperative, plurilateral approaches to raw materials governance, the EU can no longer assume continued transatlantic alignment on key issues such as supply chain diversification, open markets, and investment coordination. This development is prompting the EU to broaden its outreach and identify strategic partners beyond established frameworks such as the G7 and the MSP. While taking into consideration the capital-intensive nature of raw materials infrastructure, the section

explores the feasibility of joint financial cooperation with like-minded partners in different world regions and on a multilateral level, as well as the EU's prospects for influencing international standard-setting in the sector. The section also examines how the EU can cooperate with difficult partners such as China and the Gulf States.

Section 5 presents a brief conclusion and summarises the key policy recommendations.

1 From Extraction to Transformation: Enabling Local Value Creation

Rising geopolitical competition, growing demand for transition minerals, and increasing concerns over supply chain resilience have intensified pressure on consumer countries and companies to offer more tangible benefits in return for the reliable supply of minerals.[3] Governments and civil society actors in producing countries are demanding fairer outcomes, particularly in terms of local economic and social benefits. Value addition is widely promoted as a means to increase public revenues, create jobs, and support resource-based industrialisation.

These ambitions are not new; they reflect responses to longstanding critiques of unequal trade relations and the limited developmental outcomes many countries have experienced after decades of mineral extraction. In the past, however, efforts to actively build industries were often constrained by market-oriented governance models, especially under the influence of the Washington Consensus.[4]

Today's shifting global context has opened new political space for rethinking industrial policy in the raw materials sector, but it has also raised expectations and hopes. One key challenge lies in the conceptual ambiguity of "value addition", which is used inconsistently across actors and contexts.[5] European Union (EU) raw materials partnerships, for example, often frame value addition as a "win-win" goal and commit to supporting it in principle, but frequently there is a lack of clarity on what this entails in practice.

The case studies of Chile and Zambia help to illustrate both the political ambition and the structural diversity shaping national value addition strategies. As a high-income country and emerging market economy in Latin America, Chile has comparatively greater

[3] Thomas Scurfield, Matthieu Salomon and Silas Olan'g, *Six Keys to Unlocking Equitable Value Addition in Mining* (Natural Resource Governance Institute, January 2024), https://resourcegovernance.org/sites/default/files/2025-01/Six_Keys_Unlocking_Equitable_Value_Addition_Mining.pdf (accessed 24 June 2025).

[4] Jewellord T. Nem Singh, "The Advance of the State and the Renewal of Industrial Policy in the Age of Strategic Competition", *Third World Quarterly* 44, no. 9 (2023): 1919–1937; Tilman Altenburg, *Industrial Policy in Developing Countries: Overview and Lessons from Seven Country Cases,* Discussion Paper 4/2011 (Bonn: Deutsches Institut für Entwicklungspolitik, 2011), https://www.econstor.eu/bitstream/10419/199359/1/die-dp-2011-04.pdf (accessed 24 June 2025).

[5] Scurfield, Salomon and Olan'g, "Six Keys to Unlocking Equitable Value Addition in Mining" (see note 3); Sophia Pickles, *Value Addition in the Context of Mineral Processing* (Berlin: Heinrich Böll Foundation, 2023), https://www.boell.de/sites/default/files/2023-11/e-paper_value_addition_in_the_context_of_mineral_processing.pdf (accessed 13 August 2024).

policy space to support industrial planning.[6] Zambia, by contrast, faces tighter fiscal constraints and weaker institutional capacity as a lower-income country in Southern Africa.[7] Despite these differences, both countries are actively pursuing policies to create "linkages" and retain a greater share of value locally—highlighting not only the variation in approaches, but also the importance of context-specific conditions to achieve value addition and ultimately resource-based industrialisation.

A broader and more precise understanding of what value addition means is urgently needed: to clarify what producer countries are demanding, to define what the EU can meaningfully offer, and to guide a more realistic and differentiated approach to partnerships and industrial cooperation.

Supply vs Value Chain

The terms "supply chain" and "value chain" are sometimes used interchangeably, despite there being an important difference between them. The term "supply chain" describes the organisational and logistical connection of a (transnational) network of suppliers, manufacturers, and distributors involved in the development of a product or service. The term "value chain" goes beyond the organisational structure of the chain of activities and further describes the tangible or intangible value that is added at each stage in the development of a product or service.[8] The value generated along the value chain is not equally distributed across all stages. Traditionally, the greatest profits are generated in upstream activities, including research and development, as well as in downstream activities, primarily in sales, marketing, and services (see Figure 1).

Applied to the raw materials sector, this means that production steps with the lowest value-added share—extraction and the initial processing of raw materials—are mainly concentrated in countries of the so-called Global South, while activities with the highest profit margins are predominantly located in highly industrialised Global North countries.[9] In addition to generating less value than other segments of the value chain, the extraction of raw materials incurs additional external costs such as environmental degradation, social conflicts, and health risks.[10] This asymmetric distribution of profits and costs among actors in the value chain is often criticised, particularly in extraction countries, and it holds significant importance in current discussions surrounding new raw materials partnerships. According to a 2023 survey by the Intergovernmental Forum on Mining,

[6] Inga Carry, *Raw Materials Partner Chile: More Than Just a Supplier,* SWP Comment 2025/C 26 (Berlin: Stiftung Wissenschaft und Politik [SWP], June 2025), doi: 10.18449/2025C26.

[7] Meike Schulze, *The Strategic Raw Material Partnership Between the EU and Zambia: Industrial Cooperation Is Key to Value Creation and Long-Term Collaboration,* SWP Comment 2025/C 19 (Berlin: Stiftung Wissenschaft und Politik [SWP], May 2025), doi: 10.18449/2025C19.

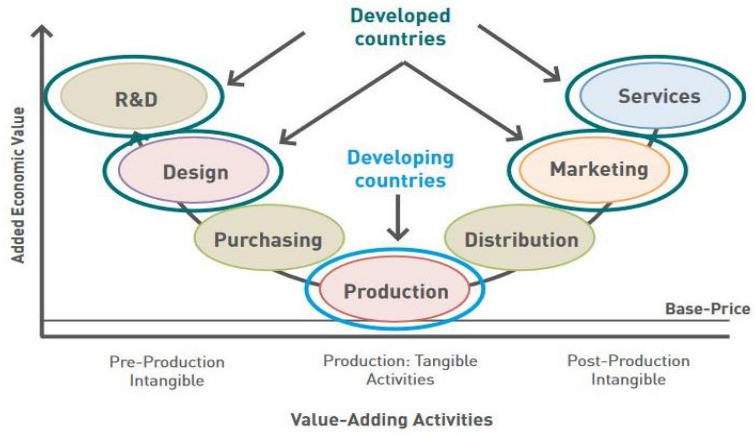
[8] Gary Gereffi and Karina Fernandez-Stark, *Global Value Chain Analysis: A Primer*, 2nd Edition (Durham, NC: Duke University, July 2016), https://dukespace.lib.duke.edu/items/e8010f4b-90c9-4835-bc54-16422bb7efb6 (accessed 19 August 2025).

[9] Ibid.

[10] Rosalie Arendt, Vanessa Bach and Matthias Finkbeiner, "The Global Environmental Costs of Mining and Processing Abiotic Raw Materials and Their Geographic Distribution", *Journal of Cleaner Production* 361 (2022): 132232; Lukas Rüttinger, Fiona Schüler, Christine Scholl and Anna Bach, *Die Deutschen Rohstoffpartnerschaften - Analyse Der Umsetzung Und Ausblick: Auftragsstudie,* RohPolPress-Kurzanalyse 6 (Berlin: adelphi, Dezember 2016), https://bit.ly/3AF7Jvp (accessed 15 August 2024); Lucia Mancini and Serenella Sala, "Social Impact Assessment in the Mining Sector: Review and Comparison of Indicators Frameworks", *Resources Policy* 57 (2018): 98–111; Adator S. Worlanyo and Li Jiangfeng, "Evaluating the Environmental and Economic Impact of Mining for Post-Mined Land Restoration and Land-Use: A Review", *Journal of Environmental Management* 279 (2021): 111623.

Minerals, Metals and Sustainable Development (IGF) conducted with 70 mineral-producing countries from Latin America, Africa, and Asia, the biggest driver of their minerals strategy was the prospect of increased beneficiation, followed by higher revenues.[11] The question of how resource-rich developing countries can achieve a higher value-added share in global raw material supply chains should therefore not only be seen as a developmental policy interest, but rather as a central component of an attractive offer made by consumer countries to producing countries, and thus a prerequisite for the successful diversification of resource partners.

Figure 1: The Smile Curve: Distribution of Value-Adding Activities along the Global Value Chain



Source: Gary Gereffi and Karina Fernandez-Stark, *Global Value Chain Analysis: A Primer*, 2nd Edition (Durham, NC: Duke University, July 2016), https://dukespace.lib.duke.edu/dspace/handle/10161/12488 (accessed 19 August 2024).

The effects of the extractive sector in resource-rich countries on overall economic development are often considered limited (sometimes even corrosive), not least because the mining industry is frequently viewed as an "enclave" sector characterised by low technology and only few significant connections to other industrial sectors of the local economy.[12] Another strand of literature suggests, however, that the advancing role of digital technologies in the extractive sector might open up new opportunities for local suppliers to enter the value chain.[13] The current race for the critical raw materials needed

[11] IGF, *Critical Minerals: National Priorities, Regional Initiatives, and Global Responsibilities: Outcomes of 2023 IGF Consultations with Member Countries*, https://www.igfmining.org/wp-content/uploads/2024/08/final-critical-minerals-national-priorities-regional-initiatives-global-responsibilities.pdf (accessed 24 June 2025).
[12] Hans W. Singer, "The Distribution of Gains Between Investing and Borrowing Countries", *The American Economic Review* 40, no. 2 (1950): 473–485; Jeffrey D. Sachs and Andrew M. Warner, *Natural Resource Abundance and Economic Growth,* Working Paper 5398 (Cambridge, MA: National Bureau of Economic Research, December 1995), https://www.nber.org/system/files/working_papers/w5398/w5398.pdf (accessed 15 August 2024); Jeffrey D. Sachs and Andrew M. Warner, "The Curse of Natural Resources", *European Economic Review* 45, 4-6 (2001): 827–838.
[13] Michiko Iizuka, Carlo Pietrobelli and Fernando Vargas, "Innovation in Mining Global Value Chains: Implications for Emerging Economies", in *Global challenges for innovation in mining industries*, ed. Alica Daly et al. (Cambridge: Cambridge University Press, 2022), 88–116; Anabel Marin, Lizbeth Navas-Alemán and Carlota Perez, "Natural Resource Industries as a Platform for the Development of Knowledge Intensive Industries", *Tijdschrift voor Economische en Sociale Geografie* 106, no. 2 (2015): 154–168; Tina Hunter, "Law and Policy Frameworks for Local Content in the Development of Petroleum Resources: Norwegian and

for the green and digital transformation is of particular importance, as it requires an efficient identification and exploitation of said resources. The need to cater to resource specificity—meaning the specific location where mineral deposits are extracted, as well as the specific characteristics of a mineral and the methods of extraction in a given environment—offers the "possibility of drawing on local skills and knowledge as well as the local presence of technology spill overs".[14] Resource-rich countries such as Norway, Sweden, Australia, and Canada are often presented as examples of how the commodities sector can create an interactive symbiosis with other industrial sectors connected through backward, forward, and horizontal linkages.[15] Morris et al. therefore suggest that "where commodity dependence is associated with a weak industrial sector, this is more often the result of the underdevelopment of the industrial sector rather than a consequence of the destructive impact of commodities production on industry".[16] The development of linkages between the commodities sector and other industries of the local economy draws on a framework developed by Hirschman, who defined three major types of linkages from the commodities sector: fiscal linkages (resource rents generated from the commodities sector and paid out to the government); consumption linkages (the demand for output of other sectors as a result of wages earned in the commodities sector); and production linkages (including forward and backward linkages).[17] The following section focuses on production linkages between the mining sector and other industrial sectors (see Figure 2).

Added Value through Upstream Linkages

Linkages of the mining sector typically exist with upstream industries, including those involved in equipment and machinery, (digital) technology, infrastructure, and research and innovation. Due to the location specificity of mineral deposits, the extraction of raw materials may take place in particularly high or deep mining areas, under specific weather conditions, as well as in an environment characterised by specific social dynamics.[18] Companies faced with these location-specific mining conditions increasingly seek input from specialised knowledge-intensive mining services providers that offer context-specific services and technologies—from exploration and mine planning services to

Australian Perspectives on Cross-Sectoral Linkages and Economic Diversification", *Mineral Economics* 27 (2014): 115–126; Mike Morris, Raphael Kaplinsky and David Kaplan, "'One Thing Leads to Another'—Commodities, Linkages and Industrial Development", *Resources Policy* 37, no. 4 (2012): 408–416; Fitsum Weldegiorgis, Evelyn Dietsche and Daniel M. Franks, "Utilizing Inter-Sectoral Linkages to Drive Economic Transformation: A Theoretical Model for the Extractive Sector", *Preprints* (2023): 2023101984.

[14] Morris, Kaplinsky and Kaplan, "'One thing leads to another'—Commodities, linkages and industrial development" (see note 13).

[15] Harold Innis, *Essays in Canadian Economic History* (Toronto: University of Toronto Press, 2018); Melville H. Watkins, "A Staple Theory of Economic Growth", *Canadian Journal of Economics and Political Science* 29, no. 2 (1963): 141–158; Gavin Wright and Jesse Czelusta, "Why Economies Slow: The Myth of the Resource Curse", *Challenge* 47, no. 2 (2004): 6–38; Magnus Blomström and Ari Kokko, "From Natural Resources to High Tech Production: The Evolution of Industrial Competitiveness in Sweden and Finland", in *Natural Resources: Neither Curse Nor Destiny*, ed. Daniel Lederman and William F. Maloney (Palo Alto, CA, Washington, DC: Stanford Economics and Finance an imprint of Stanford University Press; World Bank, 2007), 213–56.

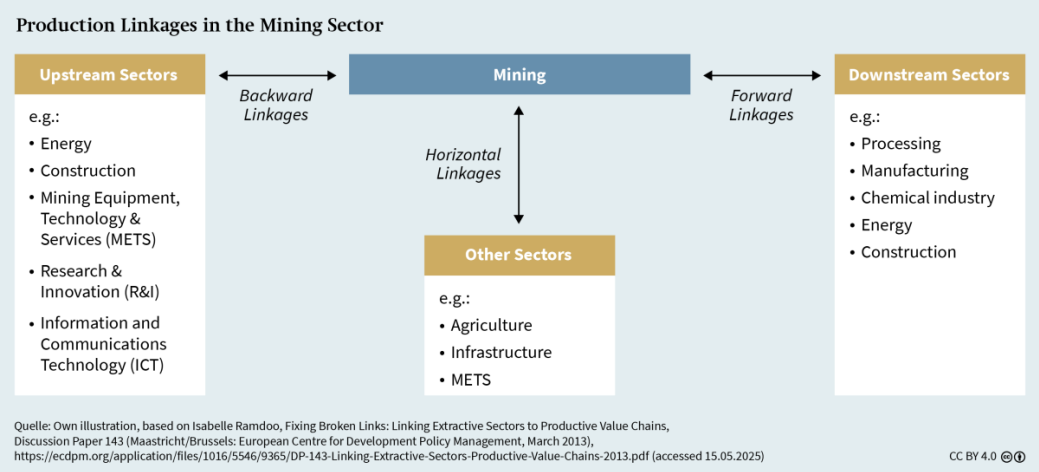
[16] Morris, Kaplinsky and Kaplan, "'One thing leads to another'—Commodities, linkages and industrial development" (see note 13).

[17] Albert O. Hirschman, *Essays in Trespassing: Economics to Politics and Beyond* (Cambridge: Cambridge University Press, 1984).

[18] Michiko Iizuka, Carlo Pietrobelli and Fernando Vargas, "Innovation in Mining Global Value Chains: Implications for Emerging Economies" (see note 13).

equipment design and environmental engineering services.[19] For example, the Chilean company Neptuno manufactures special pumps for one of the world's largest copper mines at 4,500 metres above sea level and has been Chile's first energy supplier to provide the mining industry with 100% renewable energy since 2021. Similarly, the South African engineering firm Mecad Systems provides customised mine ventilation control systems and safety solutions tailored to deep-level gold and platinum mines in Southern Africa.

Figure 2: Backward, Forward, and Horizontal Linkages between the Mining and Other Sectors



Yet, despite their location advantage, local providers face challenges, particularly in the form of high supply-chain entry barriers, skills and technology shortages, rigid supplier networks, and unequal power dynamics between lead firms and local providers.[20] Research data show that international mining companies are more inclined to rely on existing business partnerships and are therefore unwilling to risk engagement with new and unproven local providers.[21] Studies also show that mining companies often invest in R&D in extraction countries only when they need to make production more efficient due to decreasing commodity prices.[22] Consequently, it is often up to local providers to raise the necessary capital and bear the associated risks, which few local companies can afford.[23] Most important, however, is the availability of local productive capacity, that is, a

[19] Morris, Kaplinsky and Kaplan, "'One thing leads to another'—Commodities, linkages and industrial development" (see note 13); Osvaldo Urzúa, *Emergence and Development of Knowledge-Intensive Mining Services (KIMS),* Working Papers in Technology Governance and Economic Dynamics 41 (June 2012), https://www.researchgate.net/publication/254452405_Emergence_and_Development_of_Knowledge-Intensive_Mining_Services_KIMS (accessed 24 June 2025).

[20] Raphael Kaplinsky, Mike Morris and Dave Kaplan, *A Conceptual Overview to Understand Commodities, Linkages and Industrial Development in Africa,* MMCP Discussion Paper 12 (2011), https://oro.open.ac.uk/30534/ (accessed 25 June 2025); Michiko Iizuka, Carlo Pietrobelli and Fernando Vargas, "Innovation in Mining Global Value Chains: Implications for Emerging Economies" (see note 13).

[21] Carlo Pietrobelli, Anabel Marin and Jocelyn Olivari, "Innovation in Mining Value Chains: New Evidence from Latin America", *Resources Policy* 58 (2018): 1–10.

[22] Rebecca Hanlin and Christopher Hanlin, "The View from Below: 'Lock-in' and Local Procurement in the African Gold Mining Sector", *Resources Policy* 37, no. 4 (2012): 468–474.

[23] Beatriz Calzada Olvera and Michiko Iizuka, *How Does Innovation Take Place in the Mining Industry? Understanding the Logic Behind Innovation in a Changing Context* (Maastricht: Maastricht Economic and Social Research Institute on Innovation and Technology, May 2020), https://ideas.repec.org/p/unm/unumer/2020019.html (accessed 13 August 2024); Pietrobelli, Marin and Olivari, "Innovation in mining value chains" (see note 21); Patricio Meller and Pablo Parodi, *La Importancia De La Capabilidad Tecnológica En Un Mundo Global* (Santiago de Chile: Cieplan, November 2016),

skilled labour force with high levels of education and training that can supply inputs to the mining industry. Especially as technologies used in the extractive sector and upstream value chains are changing rapidly, local suppliers need to be able to adapt to and improve new technologies.[24]

Investments in this area could lead to a win-win situation for both mining companies and extraction countries. Collaboration with local providers and research institutes creates jobs and capital flows. Innovative solutions—including more efficient and environmentally friendly resource extraction methods—would not only benefit mining companies, but also preserve the resources and environment of the extraction country as well as increase the social acceptance of mining activities among local populations. Investments in mining equipment, technology, and services (METS) and green energy sources would support the energy and mobility transition in extraction countries and provide affordable and sustainable energy, especially for energy-intensive raw material processing.

According to calculations by the Natural Resource Governance Institute (NRGI), approximately 45% of the annual gross revenue generated by the raw materials sector flows into the supplier industry.[25] This represents a significant potential for strengthening local economies and value creation. To harness this potential, governments of resource-rich countries could develop targeted industrial policies[26] designed to actively integrate local suppliers into the mining sector, for example through the use of state subsidies, import protection, exemption from specific regulations, or public–private partnerships.[27] This is also the case in both Chile and Zambia (see Box 1). Often, these incentives are tied to conditions—for example, they are offered only to companies with less than a certain number of employees, require specific investments to be made by the companies, or include local content policies (LCPs)—mandating companies to use domestically supplied services or manufactured goods. The design and efficacy of LCPs, however, is contested. Weldegiorgis et al. argue, for example, that if there is a mismatch between the local productive capacity and the needs of mining companies, "mandatory LCPs with stringent targets could have a detrimental consequence on the host country".[28] They suggest that host countries should strive for a more balanced approach between LCPs and fiscal policies, while at the same time ensuring policy coherence and coordination, as well as accompanying measures to establish governance and transparency systems to prevent corruption and favouritism.[29] At the same time, LCPs are also criticised for conflicting with

https://scioteca.caf.com/bitstream/handle/123456789/1073/Capabilidad_Tecnologica_en_un_Mundo_Global%2876%29-0317.pdf (accessed 20 August 2024); Morris, Kaplinsky and Kaplan, "'One thing leads to another'—Commodities, linkages and industrial development" (see note 13).

[24] Morris, Kaplinsky and Kaplan, "'One thing leads to another'—Commodities, linkages and industrial development" (see note 13).

[25] Robert Pitman and Kaisa Toroskainen, *Beneath the Surface: The Case for Oversight of Extractive Industry Suppliers* (New York: Natural Resource Governance Institute, September 2020), https://resourcegovernance.org/sites/default/files/documents/beneath_the_surface.pdf (accessed 15 August 2024).

[26] Industrial policies can be understood as "government policies that explicitly target the transformation of the structure of economic activity […] to stimulate innovation, productivity, and economic growth." See Réka Juhász, Nathan Lane and Dani Rodrik, "The New Economics of Industrial Policy", *Annual Review of Economics*, no. 16 (2024): 213–242.

[27] Ibid.

[28] Fitsum S. Weldegiorgis, Evelyn Dietsche and Daniel M. Franks, "Building Mining's Economic Linkages: A Critical Review of Local Content Policy Theory", *Resources Policy* 74 (2021): 102312.

[29] Ibid; Gary C. Hufbauer and Jeffrey J. Schott, "Introduction: The LCR Phenomenon", in *Local Content Requirements: A Global Problem*, ed. Gary C. Hufbauer and Jeffrey J. Schott (Washington, DC, 2013), 1–16; Michael W. Hansen, Lars Buur, Anne Mette Kjær and Ole Therkildsen, "The Economics and Politics of Local

the principles of the World Trade Organization (WTO) and creating economic distortions in foreign direct investment (FDI).[30]

Added Value through Downstream Linkages

Value creation in the downstream raw materials supply chain occurs through the processing of raw materials into more refined, higher-value products—a process also referred to as beneficiation. After extraction, raw materials are typically processed and refined in smelters and refineries. In addition to the processing of primary material, some smelters and refineries may also incorporate recycling capacities of scrap and secondary material. The next stage of the downstream chain comprises the production of precursor material and the manufacturing of semi-finished and finished goods. Today, these segments of the value chain are largely concentrated in downstream markets rather than in the countries of extraction. For example, while Australia and Chile are the biggest producers of lithium, they make up only 11% and 9%, respectively, of global lithium refining. At the same time, more than 80% of battery materials comprising cathodes, anodes, electrolytes, and separators come from only three countries—China, Japan, and South Korea. Around 75% of lithium-ion cell production is currently located in China,[31] even though the share of production in other industrial countries, especially in the United States (US) and Europe, is rising continuously.

This uneven distribution has prompted growing political ambition, particularly across countries of the so-called Global South. Both Chile and Zambia illustrate this trend. Chile is pursuing efforts to expand its capacities in copper smelting and along the lithium-ion battery (LIB) supply chain, while Zambia aims to develop processing capacity for other critical minerals. In both cases, governments have expressed the longer-term goal of building battery-related manufacturing segments, where they see the highest potential to capture additional value (see Box 1).

Governments can deploy a range of industrial policy tools to support the development of downstream value chain capacity—ranging from incentives ("carrots") to regulatory measures ("sticks"). Incentive-based approaches typically aim to stimulate industrial capacity through public funding, subsidies, or tax breaks. For example, the Canadian government established the Strategic Innovation Fund that provides funding for projects aimed at enhancing Canada's capacities in processing, manufacturing, and recycling. As part of the US Inflation Reduction Act (IRA), President Joe Biden expanded the Advanced Energy Project Credit that offered tax credits for projects related to clean energy manufacturing and recycling and critical materials refining, processing, and recycling. To encourage the development of downstream industries in the lithium value chain, Chile has introduced preferential pricing schemes for companies investing in local value-adding activities, such as investing in processing or cathode manufacturing plants. Zambia, in turn, plans to introduce production-sharing agreements—targeting primarily the copper sector—to ensure domestic access to raw materials at prices below international market

Content in African Extractives: Lessons from Tanzania, Uganda and Mozambique", *Forum for Development Studies* 43, no. 2 (2016): 201–228.

[30] Camille Landais et al., *The Inflation Reduction Act: How Should the EU React?* (Paris and Wiesbaden: Franco-German Council of Economic Experts, 2023), https://www.sachverstaendigenrat-wirtschaft.de/fileadmin/dateiablage/Publikationen/FGCEE/CAE-SVG_Joint_statement_IRA_2309.pdf (accessed 13 August 2024).

[31] Jakob Kullik, *Versorgungssicherheit Mit Kritischen Rohstoffen? Analyse Der Rohstoffpolitik Deutschlands Im Europäischen Kontext* (Baden-Baden: Nomos, 2024). S. 417.

levels for local manufacturers. In addition, the country is establishing special economic zones to attract investment in mineral-based manufacturing (see Box 1).

In contrast, Indonesia has pursued a strategy of export restrictions on unrefined nickel and bauxite to stimulate investment in its processing industry. Although Indonesia's ban on unrefined nickel has largely achieved its objective of increasing processing capacity and fostering value addition—evident through job creation and higher export revenues—this progress has not come without costs. Critics point out that the rapid expansion of smelting capacity has been accompanied by a rise in greenhouse gas emissions and environmental pollution, largely due to the use of fossil fuels to power processing facilities. Additionally, as Warburton notes, Indonesia's "nickel-led growth relies on narrow networks of foreign and politically connected domestic capital, and the industry's benefits are unevenly distributed".[32] Moreover, while the export ban has been effective in boosting the nickel sector, the outcome has been quite different for bauxite. This divergence reflects Indonesia's distinct position in each market. When the ban was introduced in 2014, Indonesia was the world's largest producer of nickel ore, and key consumers—particularly China—depended heavily on its supply. In contrast, although Indonesia was the second-largest bauxite producer when the export ban was introduced, it ranked only sixth in terms of reserves. This made it easier for China to shift to alternative suppliers, and the result was simply a decline in Indonesian bauxite production, without a corresponding build-up in domestic processing.[33]

Indonesia's experience illustrates that the success of efforts to promote domestic processing capacity hinges on several key factors. Establishing smelters and refineries is capital-intensive, while the profitability varies widely depending on the specific raw material and contextual factors, including global commodity prices and operational challenges.[34] Since processing is highly energy-intensive, access to affordable, reliable, and ideally renewable energy is a critical prerequisite, especially in countries with high risks of load shedding. Reliable transport infrastructure is also essential to support the movement of raw and processed materials. Additionally, today's global raw material processing landscape is characterised by a high concentration and dominance of only a few global suppliers of refined raw materials. According to the International Energy Agency (IEA), China processes 91% of the global supply of REE, 45% of copper, 78% of cobalt, and 70% of lithium (all of which are considered critical or strategic raw materials by most industrial countries).[35] Over the past two decades, China has invested heavily in its processing capacities, aided by state subsidies for energy costs. China's approach of scaling-up processing capacities has at times resulted in excess capacity in global raw material processing markets and has therefore kept global prices for certain processed raw materials extremely low.[36]

The manufacturing of semi-finished and finished products is also subject to several contextual factors, including the availability of a skilled labour force and infrastructure, as well as proximity to the consumer market. Establishing an entire value chain, from extraction to product completion (so-called mine-to-market) in a single country is feasible

[32] Eve Warburton, "Nationalist Enclaves: Industrialising the Critical Mineral Boom in Indonesia", *The Extractive Industries and Society* 20 (2024): 101564.

[33] Scurfield, Salomon and Olan'g, "Six Keys to Unlocking Equitable Value Addition in Mining" (see note 3).

[34] Pickles, "Value Addition in the Context of Mineral Processing" (see note 5).

[35] IEA, "Global Critical Minerals Outlook 2025" (see note 1).

[36] Deutsche Rohstoffagentur in der Bundesanstalt für Geowissenschaften und Rohstoffe, *DERA-Studie Warnt Bei Silizium vor Hoher Marktkonzentration in China: Weltweite Produktionskapazitäten Übersteigen Die Nachfrage nach Silizium Deutlich*, https://www.deutsche-rohstoffagentur.de/DE/Gemeinsames/Oeffentlichkeitsarbeit/Pressemitteilungen/BGR/DERA/dera-bgr-2023-11-21_silizium-hohe-marktkonzentration-in-china.html?nn=5096114 (accessed 13 August 2024).

only in few cases. One potential candidate might be Canada, which aims to extract and process its raw materials; produce the cathodes, anodes, and batteries needed for an electric vehicle (EV); manufacture and assemble EVs; as well as recycle batteries and recover their raw materials. The decisive advantage for Canada in this regard is its well-established raw materials sector, its access to a stable and green energy supply, as well as its close integration into the US automotive supply chain.[37] Yet, competitive disadvantages continue, as manufacturing costs for semi-finished products such as anodes are significantly higher in Canada than they are in China, and current tariff policies introduced by the Trump administration could negatively impact the regional integration of supply chains. The past couple of years have also shown that, even with fiscal stimulus packets from governments, such as the IRA, countries struggle to build up certain infant industries. Although the IRA has stimulated a significant number of new projects in the mining, processing, and manufacturing sectors, many of those projects face delays or have already been halted partly due to long permitting processes, but also due to global price fluctuations, reduced competitiveness with dominant actors such as China, and economic and geopolitical uncertainty.[38]

Added Value through Horizontal Linkages

Where resource-rich countries have succeeded in building linkages from the extractive sector, those linkages are often limited to the development of (upstream and downstream) supply chains serving the mining sector. These linkages therefore contribute little to the diversification of the local economy, and instead increase economic dependency on the profitability of the extractive sector. Weldegiorgis et al. also argue that the narrow conceptualisation of LCPs regarding the development of the extractive sector and supply chains feeding into the mining industry further exacerbates this problem. Rather, "the mining sector's role within the greater economy [should not be] treated in isolation (or as a dominant contributor), but in conjunction with and in consideration of how it links with the other economic sectors and in the wider context of countries' public policies and political-economic decision making."[39]

The coupling of the raw materials sector with other areas of the economy is particularly important in light of fluctuating commodity prices and the finite nature of resources. For one thing, inter-sectoral coupling can help cushion national or local economic downturns during price slumps or after mine closures. For another, local suppliers that have built specialised knowledge, skills, and technological capacity can use horizontal linkages to enter into new and different markets.[40] For example, the exploration and extraction of metals require specialised equipment that can be used not only in mining or metallurgy, but also in other industries such as mechanical engineering. Collaboration between different sectors creates a positive feedback effect: Mining companies can increase their efficiency and profitability by tapping into expertise and technologies from companies in

[37] Inga Carry, *Critical Raw Materials Partner Canada: An (Almost) Perfect Match: The European-Canadian Raw Materials Partnership in Times of Friendshoring,* SWP Comment 2024/C 29 (Berlin: Stiftung Wissenschaft und Politik [SWP], July 2024), doi: 10.18449/2024C29.

[38] Amanda Chu, Alexandra White and Rhea Basarkar, "Delays Hit 40% of Biden's Major IRA Manufacturing Projects", *Financial Times*, 12 August 2024, https://www.ft.com/content/afb729b9-9641-42b2-97ca-93974c461c4c (accessed 14 July 2025).

[39] Weldegiorgis, Dietsche and Franks, "Utilizing Inter-Sectoral Linkages to Drive Economic Transformation: A Theoretical Model for the Extractive Sector", p. 9 (see note 13).

[40] Morris, Kaplinsky and Kaplan, "'One thing leads to another'—Commodities, linkages and industrial development" (see note 13).

other sectors. These companies leverage the existing demand to invest in the further development of their products and services.

Additionally, inter-sectoral coupling contributes to reducing infrastructure redundancies, provided that infrastructure (both soft and hard) is not enclave-oriented—that is, built solely for the needs and purposes of the extractive industry—but available and linked to other sectors and economic actors.[41] As part of Canada's Critical Minerals Strategy, the government established the Critical Minerals Infrastructure Fund that provides federal funding for transport and clean energy for projects related to the development of critical minerals. Additionally, in many African and Latin American countries, coupling the raw materials sector with the agriculture sector also offers opportunities. Particularly when mining and agriculture coexist geographically, synergies can be created and potential conflicts between the two sectors mitigated. For instance, expanding (sustainable) energy sources and roads, which both sectors can use, can prevent infrastructure redundancies. Innovations such as improvements in water use and treatment in mines can positively influence irrigation methods in agriculture. Simultaneously, this approach can help reduce the negative environmental impacts of mining on surrounding cultivated areas. Horizontal linkages with the energy and mobility sectors are particularly prudent, as demand for the raw materials that are used for the manufacturing of green energy sources and EVs continues to rise. Countries endowed with the mineral resources needed for the green energy and mobility transition should therefore leverage the intersection of both sectors, not merely for the sake of economic development, but also to advance their own green and digital transition.

However, the practical facilitation of horizontal linkages is often challenging and requires intensive policy planning and long-term strategic coordination. The most direct and frequently observed connections tend to be with the energy and transport sectors, which have a very clear link to mining operations. These linkages, however, are often limited in terms of technological exchange or broader industrial integration. In Chile, promising efforts are being made to couple mining with the development of green hydrogen technologies. In Zambia, by contrast, horizontal linkages are not prominent yet in policy planning (see the following Box 1).

Box 1: Value Addition and Sector Linkages: Strategies of Chile and Zambia

Chile: Partnerships for Value Creation and Downstream Development

Strategic Frameworks:

- National Lithium Strategy (2023)
- National Strategy for Strengthening Smelting and Refining Capacity (2023)
- Critical Minerals Strategy (in development as of May 2025)

Sector Objectives:

Chile is the world's largest producer of copper and the second-largest producer of lithium—both designated as "*critical raw materials*" by the EU and several other industrialised economies. While Chile aims to expand production of both minerals, it also seeks to

[41] Ibid; Adeolu O. Adewuyi and T. A. Oyejide, "Determinants of Backward Linkages of Oil and Gas Industry in the Nigerian Economy", *Resources Policy* 37, no. 4 (2012): 452–460; Zeferino Teka, "Linkages to Manufacturing in the Resource Sector: The Case of the Angolan Oil and Gas Industry", *Resources Policy* 37, no. 4 (2012): 461–467.

strengthen the role of the state in the mining sector, enhance smelting and refining capacity, and build local value addition through downstream industrial development.

Key Policy Components:

- Public–private partnerships for lithium extraction in strategic salt flat, with majority ownership by state-owned enterprises
- Preferential pricing models: Companies can access lithium at reduced rates if they commit to domestic downstream value addition
- Expanding and modernising smelting and processing capacities in the copper sector
- R&D support focused on sustainable lithium production and the (re)use of batteries

Upstream:

- Local supplier development: Programmes such as the World Class Supplier Program foster collaboration between lead firms and local suppliers through skills training and technology transfer
- Exploration and R&D: The state supports the development of sustainable exploration and extraction techniques via a national research institute

Downstream:

- Processing: Expansion and modernisation of smelting and refining capacities through public–private collaboration; special lithium operation contracts are given primarily to companies committed to downstream local value addition
- Manufacturing: Preferential pricing mechanisms aim to attract the establishment of downstream industries, for example battery refining or cathode plants

Horizontal:

- Energy sector: Expansion of renewable energy sources and battery storage; long-term power purchase agreements between mining companies and renewable energy providers
- Water sector: Investment in desalination plants; promotion of closed-loop systems
- Hydrogen sector: Development of a green hydrogen economy for both export and domestic use, especially in the mining sector

Challenges:

- Balancing production expansion with ecological concerns and the rights and interests of local and indigenous communities
- Structural disadvantages for downstream manufacturing due to Chile's distance to major consumer markets, especially in the EV supply chain
- Local suppliers struggle to compete on international markets

Zambia: Fostering Investment and State Participation

Strategic Frameworks:

- 8th development plan 2022-2026
- National Mineral Resource Development Policy (2022)
- National 3 million Tonnes Cooper Production Strategy by 2031 (2024)
- National Critical Minerals Strategy (2024)

Sector Objectives:

Zambia's raw materials strategy (updated in 2024) places primary emphasis on expanding mineral extraction, while aiming to secure greater domestic value creation through increased state participation in mining operations. Although downstream processing and regional industrial development are long-term goals, they have received comparatively limited practical focus to date.

Key Policy Components:

- Modernisation of geological data and exploration initiatives
- Expansion of the copper sector and diversification of mineral extraction and exports
- Introduction of a "free equity model" allowing up to 30% state ownership (via Zambia Minerals Investment Corporation) in new mining projects
- State engagement in metal trading (joint venture with Swiss company Mercuria)

Upstream:

- **Energy infrastructure:** Expansion and stabilisation of electricity supply, with a shift towards solar and coal to complement hydropower
- **Local supplier development:** Implementation of local content requirements to support domestic companies and service providers
- **Exploration and R&D:** Nationally funded exploration; university research and training programmes

Downstream:

- **Processing and beneficiation:** Concrete targets for expanded copper smelting capacity; vague and non-specific plans for the beneficiation of other minerals
- Manufacturing: Introduction of production-sharing agreements for mineral products; policy focus on copper-based manufacturing (e.g. wires, cables); use of special economic zones to attract investment
- **Battery cluster:** MoU with Democratic Republic of the Congo (DRC) (2022) on joint precursor/component production

Horizontal:

- Transport infrastructure: Expansion of key corridors (rail, road)
- Cross-sector integration: Policy aims but no concrete measures to support linkages with other sectors

Challenges:

- Balancing production expansion with ecological risks, and diverse impacts on local livelihoods
- Significant investment needs; however, high-risk perceptions and poor credit ratings deter private capital
- Policy instability and limited institutional capacity for industrial planning, including regional cooperation with neighbouring countries

Conditions for Local Value Addition in the Mining Sector

The development of production linkages between the mining sector and other segments of the domestic economy can serve as a driver of resource-based development. In particular, the establishment of intersectoral linkages facilitates economic diversification by embedding mining activities within broader value chains.[42] This approach aligns with the interests of resource-rich countries seeking to move beyond an "extractivist" model, whereby resources are merely extracted and exported while most of the beneficiation and value adding takes place elsewhere.[43]

At the same time, the potential for building production linkages hinges on a number of key factors that go beyond the mere availability of mineral deposits (see Table 1). The success of domestic raw material processing is closely tied to the availability of, and access to, reliable and affordable energy, as well as efficient transport, water, and communication infrastructure. Reliance on an unstable fossil-fuel-based energy supply introduces operational risks, adds environmental burdens, and undermines global decarbonisation objectives.[44] By contrast, investing in a resilient and, crucially, sustainable energy system supports industrial development while reinforcing commitments to climate goals.[45] This becomes even more significant in light of emerging carbon-pricing instruments, such as the EU's Carbon Border Adjustment Mechanism (CBAM), which incorporates the carbon emissions associated with the production of goods into their cost upon export to the EU. Should CBAM eventually extend to cover critical raw materials such as copper, lithium, or nickel, countries that have invested in low-emission production processes would gain a competitive edge through reduced carbon-related costs and improved market access.

Another factor influencing where value chain activities can be expanded is how well a country is integrated into regional and global production networks, along with the size of its domestic market and its proximity to major consumer regions. Many ASEAN countries benefit from their geographic proximity to major industrial hubs such as China, Japan, and South Korea, which are global leaders in EV manufacturing and battery technology.[46] For example, countries such as Thailand, Vietnam, and the Philippines have now become important hubs for automotive and electronics value chains. These economies tend to have high levels of backward participation in global value chains, meaning they import

[42] Isabelle Ramdoo, *Resource-Based Industrialisation in Africa: Optimising Linkages and Value Chains in the Extractive Sector,* Discussion Paper 179 (ECDPM, September 2015), https://ecdpm.org/work/resource-based-industrialisation-in-africa-optimising-linkages-and-value-chains-in-the-extractive-sector (accessed 14 July 2025); Weldegiorgis, Dietsche and Franks, "Utilizing Inter-Sectoral Linkages to Drive Economic Transformation: A Theoretical Model for the Extractive Sector" (see note 13).

[43] Hannes Warnecke-Berger, Hans-Jürgen Burchardt and Rachid Ouaissa, *Natural Resources, Raw Materials, and Extractivism: The Dark Side of Sustainability,* Extractivism Policy Brief 1 (2022), https://extractivism.de/wp-content/uploads/2022/01/extractivism_policy_brief-1-2022-1.pdf (accessed 27 June 2025).

[44] See, for example, Lauri Myllyvirta et al., *Debunking the Value-Added Myth in Nickel Downstream Industry: Economic and Health Impact of Nickel Industry in Central Sulawesi, Southeast Sulawesi, and North Maluku* (Centre for Research on Energy and Clean Air, 26 February 2024), https://shorturl.at/yeGwR (accessed 14 July 2025).

[45] Galina Alova, *Integrating Renewables in Mining: Review of Business Models and Policy Implications,* OECD Development Policy Papers 4 (OECD, November 2018), https://www.oecd.org/en/publications/integrating-renewables-in-mining_5bbcdeac-en.html (accessed 14 July 2025).

[46] Younkyoo Kim, *U.S.-China Electric Vehicle Battery Competition and the Role of South Korea,* Congressional Affairs (NBR, December 2023), https://www.nbr.org/wp-content/uploads/pdfs/publications/brief-younkyoo-kim-dec23.pdf (accessed 14 July 2025).

intermediate goods, add value, and then export finished or semi-finished products.[47] This global value chain structure supports further industrial upgrading, including the development of downstream capabilities. By contrast, resource-rich countries in Africa and Latin America are more often positioned at the start of the value chain, exporting raw materials such as lithium, copper, and cobalt with limited domestic transformation. Their trade patterns are more heavily characterised by forward participation, meaning their exports are used in the production of goods elsewhere and often without strong linkages to local or regional processing industries.[48] This is compounded by geographical distance from major green tech markets, which can create additional barriers to developing downstream capabilities.

Table 1: Conditions for Local Value Addition in the Mining and Downstream Industries

- Quantity, quality, and recovery rate of mineral reserves
- Global concentration of mineral reserves and production capacity
- Access to reliable, competitively priced, and sustainable hard infrastructure (energy, water, transport, communication)
- Availability of soft infrastructure (skilled labour force, education and R&D, social infrastructure, and community engagement)
- Proximity to end market
- Governance, political and economic stability, and investment climate
- Political and trade partnerships (free trade agreements (FTAs), strategic partnerships, etc.)

Source: Authors' own illustration.

One of the greatest hurdles remaining for the development of the raw materials sector is access to the necessary capital and financing. Since the mining sector is particularly capital- and risk-intensive and subject to price fluctuations, its development hinges on a favourable investment climate and access to (public) seed funding. In many resource-rich countries, the mining sector is supported by a mix of public and private funding. This is sensible when financial frameworks are designed to ensure that companies benefiting from mining or processing also contribute to regional development and value creation without relieving the state of fundamental responsibilities such as public infrastructure development. The investment climate of a country or region is inherently shaped by the stability of the political and economic environment, as well as its institutional capacities, or its "social" infrastructure, meaning the efficiency of the administrative and regulatory regime.[49] Political and economic stability is critical, both for attracting FDI as well as for the government's ability to pursue long-term developmental strategies.

Political and trade partnerships also shape a country's potential to develop downstream industries. FTAs with major consumer markets can facilitate the exchange of

[47] Asian Development Bank, *ASEAN and Global Value Chains: Locking in Resilience and Sustainability* (Metro Manila: March 2023), https://www.adb.org/sites/default/files/publication/871976/asean-global-value-chains-resilience-sustainability.pdf (accessed 14 July 2025). doi: 10.22617/SPR230100-2.
[48] Christine Z. Qiang, Yan Liu and Victor Steenbergen, *Chapter 1: An Investment Perspective on Global Value Chains* (World Bank Group, December 2021), https://shorturl.at/yEIX2 (accessed 14 July 2025).
[49] Morris, Kaplinsky and Kaplan, "'One thing leads to another'—Commodities, linkages and industrial development" (see note 13).

minerals and manufactured goods and may offer additional incentives.[50] For example, under the IRA, tax credits are offered to consumers purchasing electric or fuel cell vehicles—provided that the battery materials are sourced from countries with which the US maintains an FTA. As a result, lithium and other critical inputs from Chile qualify for these benefits, thereby incentivising the export of battery materials and potentially semi-finished products to the US market.[51] At the same time, political partnerships and MoUs can play an enabling role, but they are not in themselves sufficient to attract investments or build up downstream industries. As this paper demonstrates, raw materials partnerships such as those struck between the EU and resource-rich countries create a political framework with the potential to facilitate technical and industrial cooperation. However, the effectiveness of these frameworks ultimately depends on private-sector engagement. Investment decisions by firms will hinge far more on the factors outlined in this section than on the mere existence of bilateral agreements.

Recommendations

- **Promote local value through linkages:** Generating local value through upstream, downstream, and horizontal linkages is essential for resource-based development. In particular, horizontal linkages foster economic diversification, reduce the enclave character of the mining sector, and lessen dependency on extractive revenues.
- **Assess context-specific potential:** Resource-rich countries should identify which linkages offer the most strategic potential in their specific national contexts. Factors such as infrastructure availability, market access, reserve quality, and institutional capacity should guide this assessment. Linkage strategies should be developed through a multi-stakeholder process involving policymakers, industry, and civil society, and include clear mechanisms and incentives for vertical and horizontal integration.
- **Implement local content policies with robust governance:** LCPs can be an effective tool to promote domestic participation in mining value chains. However, their success depends on the presence of transparent, well-resourced governance mechanisms that ensure effective implementation, monitor compliance, and safeguard against corruption.
- **Invest in education, skills, and technological capacity:** Building a strong domestic workforce requires sustained investment in education and vocational training. Efforts should extend beyond mining-specific curricula to include transferable skills for a broader industrial base, with a focus on institutions that foster technological development and innovation, especially as mining becomes more automated and tech-driven.
- **Leverage political and trade partnerships:** Political and trade agreements can provide a political and legal framework for increased cooperation, including industrial and technological collaboration. However, to translate these frameworks into concrete projects, both parties must develop **a** realistic understanding of mutual

[50] Transport & Environment, *Strike a Balance: Trade Agreements for Resilient and Responsible Supply Chains: Five Recommendations for Trade Policy-Makers*, 25 June 2024, https://shorturl.at/TZUbw (accessed 27 June 2025).
[51] Reuters, "Chilean Lithium to Be Eligible for U.S. Tax Break", 11 July 2024, https://shorturl.at/N1gFV (accessed 14 July 2025); América Economía, *Chile Will Access the United States IRA Law with Its Lithium Products*, 12.07.2024, https://www.americaeconomia.com/en/node/287729 (accessed 14 July 2025).

capacities and constraints. This includes evaluating what consumer countries can offer in terms of industrial cooperation (including private-sector support), as well as assessing whether the host country's conditions can support a compelling business case for downstream investment, or where support is needed.

2 Strategies to Promote Local Value Addition in Latin America and Africa

A significant share of the world's critical mineral deposits is concentrated in Latin America, Africa, and South East Asia. Many countries in these regions are already important producers of these minerals and will play a key role in meeting global demand. Africa holds key base metal deposits such as copper and bauxite, and it accounts for an estimated 19% of global reserves of battery-relevant metals such as cobalt and lithium. Although Latin America is renowned for its copper and lithium reserves, it could also gain prominence in extracting graphite, manganese, and REEs.[52] South East Asia is equally well-positioned, with Indonesia as the world's largest nickel producer and Myanmar playing a vital role in the supply of heavy rare earths.

Although South East Asia is becoming an increasingly important player in global mineral supply chains, this working paper focuses on mineral-rich countries in the Southern African Development Community (SADC) and the South American Andes with a particular spotlight on Zambia and Chile. The two regions were selected because of their strategic position in the supply of critical minerals and their active efforts in reforming their minerals sector policies to promote local value addition. In addition, both regions have become central to the EU's raw materials strategy, as reflected in the its raw materials partnerships with Chile and Argentina in the Andes, and with the DRC, Zambia, and Namibia in the SADC region.

As the US, the EU, and other G7 states increasingly employ state intervention to secure mineral supplies and bolster domestic industries, mineral-rich countries are also intensifying efforts to promote resource-based development and foster local value addition. Beyond national strategies, initiatives for regional cooperation are also becoming more prominent in shaping resource governance, although the level of regional collaboration varies significantly between Africa and Latin America. Each region faces distinct challenges and levels of integration into global supply chains, affecting their ability to generate local value from mineral resources.

[52] Papa D. Diene, David Manley, Silas Olan'g and Thomas Scurfield, *Triple Win: How Mining Can Benefit Africa's Citizens, Their Environment and the Energy Transition* (Natural Resource Governance Institute, November 2022), https://resourcegovernance.org/publications/triple-win-mining-africa-environment-energy-transition (accessed 14 August 2024); U.S. Geological Survey, *Mineral Commodity Summaries 2024* (Reston, Virginia: U.S. Geological Survey, 2024), https://pubs.usgs.gov/publication/mcs2024 (accessed 14 August 2024); Milan Grohol and Constanze Veeh, *Study on the Critical Raw Materials for the EU 2023: Final Report* (Luxembourg: Publication Office of the European Union, 2023), https://op.europa.eu/en/publication-detail/-/publication/57318397-fdd4-11ed-a05c-01aa75ed71a1 (accessed 14 August 2024).

Africa

In most African countries, the raw materials sector remains largely disconnected from the broader economy, with only limited upstream and downstream linkages.[53] Processing and refining capacities are still underdeveloped across the continent. In the Economic Community of West African States (ECOWAS) region, few minerals—primarily gold—are processed locally, whereas the SADC region shows somewhat greater capacity, particularly in countries such as Zambia and the DRC, and more significantly in South Africa.[54]

Nonetheless, major challenges persist regarding the enabling conditions for strengthening value addition and promoting resource-based industrialisation across Africa—though with significant variations between countries (see discussion in Section 1). Key constraints are related to the continent's infrastructure deficits. Energy insecurity, water scarcity, and the growing impacts of climate change pose major obstacles to energy- and water-intensive activities such as metal processing and other types of industrial development. Another major barrier is the intense global competition in the beneficiation of minerals, and even higher-value segments of the value chain, especially in battery production. This challenge is exacerbated by China's dominant production capacity as well as its increasingly active industrial policy in industrialised economies, which restrict market access and limit policy space for emerging producers.[55] Nonetheless, new policy frameworks are being developed at both the continental level—through the African Union (AU)—and within several Regional Economic Communities (RECs), reflecting the growing momentum to promote value addition and industrial development around critical minerals.

Strategies to strengthen regional cooperation have been in place for some time. The central continental strategy, introduced in 2009, focusing on promoting local value addition in the mining sector through linkages is the Africa Mining Vision (AMV). To implement the AMV and the corresponding Action Plan, the African Minerals Development Centre (AMDC) was established in 2011, supported by the UN Economic Commission for Africa (UNECA) in the first instance, before it became an official agency of the AU in 2019. However, the AMDC is still not fully operational, as it is still missing the required 15 ratifications of its statute by AU member states.

Increased geopolitical competition for access to raw materials has injected new life into regional cooperation efforts. As a result, there has been significant progress with the implementation of planned instruments of the AMV, such as the Geological and Minerals Information System, the African Minerals and Energy Resource Classification Management System, as well as the Pan-African Reporting Code. Key players at the regional level are the AMDC as well as the African Development Bank (AfDB) via its African Natural

[53] Andrew Bowman, Tomas Frederiksen, Deborah F. Bryceson, John Childs, Emma Gilberthorpe and Susan Newman, "Mining in Africa After the Supercycle: New Directions and Geographies", *Area* 53, no. 4 (2021): 647–658.

[54] Carol Zammit, *Case Studies Report* (AfricaMaVal, 31 January 2024), https://acrobat.adobe.com/id/urn:aaid:sc:EU:5990eb9d-921b-4563-ae6a-f9f5b10609f5 (accessed 13 August 2024).

[55] Gaylor Montmasson-Chair, Lesego Moshikaro and Lerato Monaisa, *Opportunities to Develop the Lithium-Ion Battery Value Chain in South Africa* (Pretoria: Trade & Industrial Policy Strategies, March 2021), https://www.tips.org.za/images/Battery_Manufacturing_value_chain_study_main_report_March_2021.pdf (accessed 12 August 2024); Antonio Andreoni and Elvis Avenyo, *Critical Minerals and Routes to Diversification in Africa: Opportunities for Diversification into Batteries and Fuel Cells and Mining Equipment Technologies - the Case of South Africa*, Background paper commissioned by the UNCTAD secretariat for the 2023 edition of the Economic Development in Africa Report (August 2023), https://unctad.org/system/files/non-official-document/edar2023_BP3_en.pdf (accessed 13 August 2024).

Resource Centre. These two institutes developed the African Green Minerals Strategy,[56] which was finally launched in December 2024. It demonstrates ways for African countries to harness mineral resources for sustainable development, focusing on the minerals that are essential for the energy transition. Key pillars of the strategy are attracting private-sector investments and developing mineral value chains to foster industrialisation and mineral stewardship. Particularly with regard to regional value chain development and value addition, the African Continental Free Trade Area (AfCFTA) is expected to play a crucial role. However, negotiations around mineral trade still lacks a coordinated strategy for the African continent.[57]

Regional organisations such as SADC and ECOWAS have also launched their own initiatives to strengthen the raw materials sector. In ECOWAS, regional strategies such as the ECOWAS Mineral Development Policy (2011) focus on harmonising mining laws, implementing local content requirements, and governing the raw materials sector. The ECOWAS Strategic Minerals Feedstock and Input Development Strategy (first version in 2022) is currently the first strategy specifically focused on local value addition. In the SADC region, several strategies and plans aimed at increasing local value addition in the mining sector exist, including the SADC Regional Mining Vision, the Regional Indicative Strategic Development Plan, and the Industrialisation Strategy and Roadmap (2015–2063).

Efforts to foster regional integration are driven by an interest to strengthen value addition linked to local mineral resources through intensified cross-border collaboration. Some African countries support the local establishment of supply chain segments for solar cells and wind turbines, though this is mostly tied to national energy policies and only indirectly connected to the raw materials sector.[58] More strongly in focus are regional "battery clusters" in Western as well as Southern Africa. Particularly Southern Africa offers [59] favourable conditions due to its mineral deposits, mining sites, and scalable refining capacities. Feasibility studies indicate that integrating multiple processes (mining, refining, and intermediate product manufacturing) is cost-effective in the region.[60] Such integration would not only enhance local value addition, but also increase the likelihood of long-term supply contracts with global battery and automotive manufacturers.

[56] African Natural Resources Management and Investment Center (ANRC), *Approach Paper Towards Preparation of an African Green Minerals Strategy* (Côte d'Ivoire: African Development Bank Group, December 2022), https://www.afdb.org/sites/default/files/documents/publications/approach_paper_towards_preparation_of_an_african_green_minerals_strategy.pdf (accessed 27 June 2025).

[57] Marvellous Ngundu and Julia Baum, *Africa Has Critical Minerals but Needs a Unified Strategy*, 22 May 2025, https://futures.issafrica.org/blog/2025/Africa-has-critical-minerals-but-needs-a-unified-strategy (accessed 27 June 2025).

[58] Aleksandra N. Wojewska and Bernhard Tröster, *Industrial Policies for the Development of Lithium-Ion Batteries (LIB) and Renewable Energy Technology Value Chains in Africa*, April 2023 (unpublished work contract on behalf of SWP).

[59] For an overview of the strategies as of April 2023, refer to Inga Carry, Melanie Müller and Meike Schulze, *Elemente Einer Nachhaltigen Rohstoffaußenpolitik: Partnerschaften Für Lokale Wertschöpfung in Mineralischen Lieferketten,* SWP Working Paper AP 01/2023 (Berlin: 1 June 2023), https://www.swp-berlin.org/publications/products/arbeitspapiere/AP_01-2023-FG06-Rohstoffau%C3%9Fenpolitik.pdf#page=35.

[60] Edem Foli, *SADC E-Mobility Outlook: Accelerating the Battery Manufacturing Value Chain,* Occasional Paper 316 (Johannesburg: South African Institute of International Affairs, December 2020), https://saiia.org.za/wp-content/uploads/2020/12/Occasional-Paper-316-foli.pdf (accessed 19 August 2024); Deon Cloete, *SADC Futures of Mining: Implications of Large-Scale EV Adoption,* Policy Insights 94 (Johannesburg: South African Institute of International Affairs, October 2020), https://saiia.org.za/wp-content/uploads/2020/11/Policy-Insights-94-cloete.pdf (accessed 19 August 2024).

A concrete binational approach exists in the form of the "DRC-Zambia Battery Council", which was established via a cooperation agreement in April 2022. Together, Zambia and the DRC possess more than 80% of global cobalt reserves. The countries aim to establish a supply chain for batteries regionally. The project is supported by different international partners: For example, UNECA and the African Export-Import Bank supported an initial feasibility study. In 2023, the Biden administration in the US had pledged support for the project, as well as the EU in their bilateral MoUs with Zambia and the DRC.

However, the implementation of this and other regional projects has encountered setbacks. In the case of the DRC–Zambia Battery Initiative, progress has been slow due to the lack of a coordinated political commitment at the regional level and the absence of a coherent industrial strategy to guide investment and implementation. Although bilateral political dialogue continues, disagreements—such as over the location of processing facilities—have stalled further progress.[61] Moreover, despite regular bilateral political dialogues, progress with implementation has been slow, with disagreements arising, for example, over the location of processing facilities. The intensification of conflict in the DRC and the broader regional instability further complicate cooperation.

Although deeper regional integration of the battery cluster development was endorsed at the SADC summit in March 2023, problems remain in practice. Moreover, South Africa has so far played a limited role in regional battery initiatives, despite its strategic position as a hub for refining, finance, and trade.[62] But recent developments indicate that South Africa intends to expand its capacities. In early June 2025, South Africa's government released its Critical Minerals and Metals Strategy, designed to modernise the nation's current mineral policy frameworks in response to recent changes in the global critical minerals sector.[63] South Africa's industrial policy regarding critical minerals prioritises in-country processing and upgrading over the export of raw materials. The country aims to establish itself as a leading smelting and refining centre for the SADC region, but it will have to work on the implementation of this strategy.[64]

In parallel, rising protectionism in major markets—from China to the US and the EU—is making export and investment conditions increasingly challenging for African producers. Donor support for regional integration and industrial development is also declining (see also discussion in Section 4). This has constrained progress with initiatives such as the battery cluster in Southern Africa.

At the same time, growing nationalist tendencies in several mineral-rich African countries are becoming more pronounced, particularly through the increasing use of export bans and other restrictive measures. These developments ultimately hinder regional coordination. In countries such as Namibia, Zambia, the DRC, Ghana, and Uganda, there has been a visible shift away from liberal mining policies. Rather than relying on FDI and investor-friendly tax incentives, these governments are prioritising national-level

[61] Schulze, "The Strategic Raw Material Partnership between the EU and Zambia" (see note 7); Silas Olan'g and Thomas Scurfield, *The DRC-Zambia Battery Plant: Key Considerations for Governments in 2024,* Briefing (Natural Resource Governance Institute, December 2023), https://resourcegovernance.org/publications/drc-zambia-battery-plant-key-considerations-governments-2024 (accessed 19 August 2024).

[62] Olan'g and Scurfield, "The DRC-Zambia Battery Plant: Key Considerations for Governments in 2024" (see note 61).

[63] Department of Mineral and Petroleum Resources of the Republic of South Africa, *Critical Minerals and Metals Strategy South Africa* (2025), https://www.gov.za/sites/default/files/gcis_document/202505/critical-minerals-and-metals-strategy-south-africa-2025.pdf (accessed 27 June 2025).

[64] Azwimpheleli Langalanga, Melanie Müller and Adrian Joseph, *South Africa and the EU Should Collaborate to Build Strategic Interdependence in Critical Minerals,* Explainer (South African Institute of International Affairs [SAIIA], 30 June 2025), https://saiia.org.za/research/south-africa-and-the-eu-should-collaborate-to-build-strategic-interdependence-in-critical-minerals/ (accessed 30 June 2025).

local content requirements—especially in the upstream segments of the value chain.[65] Since 2021, many of these governments have introduced new export restrictions, including bans on the export of unprocessed raw materials and export duties.[66]

Latin America

In many Latin American countries, dependence on resource extraction for economic development has prioritised the export of unrefined raw materials over domestic industrialisation, leaving countries vulnerable to commodity price fluctuations and with little economic diversification. Efforts to create upstream linkages with the local supplier industries have been most successful in Chile and Brazil, where supplier development programmes—often led by big mining companies and development agencies—have helped enhance the capabilities of local suppliers and connect them with other players of the supply chain.[67] In contrast, the integration of Peruvian suppliers in the copper supply chain remains relatively low, partly due to Peru's weak innovation system and fragile local institutional setting.[68]

The integration of downstream industries also varies significantly across Latin American mining countries. In Chile, both copper and lithium are processed domestically, although the export of copper cathodes has been steadily declining, giving way to a higher share of exports of less refined copper concentrate instead; in Peru, meanwhile, very little of the copper extracted is processed domestically. Although a 2019 study found that linkages to Colombia's mining sector had actually decreased over the years, as the priority lay on revenue generation rather than economic diversification, the Colombian government has taken steps to use the rising demand for critical minerals as a new impetus for its mining industry.[69] This is indicative of a broader regional trend, with countries rich in lithium and other battery materials viewing the transition to electric

[65] Numerous studies indicate that the effectiveness of local content requirements in the region is rather low. This weak impact is likely due to the limited capacity of local suppliers and various political factors. However, there is potential for improvement if policy planning is strengthened or if multinational mining companies are actively involved in projects. See Patricia Ackah-Baidoo, "Implementing Local Content Under the Africa Mining Vision: An Achievable Outcome?", *Canadian Journal of Development Studies* 41, no. 3 (2020): 486–503; Hansen et al., "The Economics and Politics of Local Content in African Extractives: Lessons from Tanzania, Uganda and Mozambique" (see note 29); Cindy Wilhelm, "Local Content Is Politics': An Examination of the Origins of Local Content Policies in Guinea's Mining Sector", *The Extractive Industries and Society* 13 (2023): 101158.

[66] Theophilus Acheampong, *From Mines to Markets: How Africa and Europe Can Become Green Industry Partners of Choice* (London: European Council on Foreign Relations, 24 April 2024), https://ecfr.eu/publication/from-mines-to-markets-how-africa-and-europe-can-become-green-industry-partners-of-choice/ (accessed 19 August 2024).

[67] Penny Bamber, Karina Fernandez-Stark, Vivian Couto and Dayne Hamrick, *Chilean Services Export Strategy: Capitalizing on Competitive Advantages to Boost Services Exports* (Subsecretaría de Relaciones Económicas Internationales, Gobierno de Chile, November 2021), https://storage.googleapis.com/wzukusers/user-35057030/documents/8deea822991e419c981e7c39e96cc041/2021_12_17_GereffiLLC_Chile%20Services%20Export%20Strategy_ENG.pdf (accessed 14 July 2025); Carry, "Raw Materials Partner Chile: More Than Just a Supplier" (see note 6); Karina Fernandez-Stark and Penny Bamber, *Innovation and Competitiveness in the Copper Mining Global Value Chain: Developing Local Suppliers in Peru,* Discussion Paper IDB-DP-855 (Inter-American Development Bank, December 2021), https://publications.iadb.org/en/innovation-and-competitiveness-copper-mining-global-value-chain-developing-local-suppliers-peru (accessed 13 August 2024).

[68] Fernandez-Stark and Bamber, "Innovation and Competitiveness in the Copper Mining Global Value Chain: Developing Local Suppliers in Peru" (see note 67).

[69] Columbia Center on Sustainable Investment, *Linkages to the Mining Sector in Colombia* (New York: Columbia University, October 2019), https://ssrn.com/abstract=3678388 (accessed 25 June 2025).

mobility as a new opportunity to build capabilities within and beyond the mining sector, particularly along the LIB and EV supply chains.

Regional cooperation on raw material supply chains has been less pronounced in Latin America and the Caribbean (LAC) as compared to Africa. No overarching official strategies similar to the AMV have been created so far. Still, the Latin American Mining Organization, a regional multi-stakeholder body, supports regular intergovernmental meetings such as the Conference of Mining Ministries of the Americas Conference, which also invites members from Central and North America. The Economic Commission for Latin America and the Caribbean (ECLAC/CEPAL), the largest multilateral organisation in the region, works on matters related to critical raw materials through its divisions on natural resource management and economic development. In 2022, at the request of Bolivia, Chile, and Argentina, it established a Permanent Technical Dialogue Forum on Lithium Innovation, Technological Development, and Value Addition that now also includes Mexico.[70] Additionally, the organisation regularly hosts conferences and commissions studies related to the region's potential in the areas of critical raw materials, natural resource governance, the energy transition, and sustainable economic, ecological, and social development. In 2024, one of those studies explored the potential and challenges of regional cooperation in the lithium supply chain and argues that the rising demand for battery materials such as lithium could serve as a driver for economic diversification and regional collaboration.[71] The authors identify several areas of action including a) scientific and technical cooperation comprising joint research on more efficient and environmentally friendly extraction methods such as Direct Lithium Extraction, geological surveying, and land and water governance; b) industrial collaboration, especially in the form of battery clusters and the production of cathodes, anodes, electrolytes, and separators; and c) increased coordination on financing through multilateral development banks.

In the field of scientific and technical cooperation, several initiatives have been launched in recent years. In addition to the Technical Dialogue Forum, Chile has hosted the Foro del Litio (Lithium Forum) since 2023, bringing together industry representatives and political decision-makers from across the Lithium Triangle. In November 2023, Chile and Argentina also signed an MoU to establish a Binational Lithium and Salt Flats Working Group.

Stronger regional integration of both the raw materials and downstream segments of the lithium supply chain could generate important synergies and build on the complementary strengths of each country. For example, while Chile and Argentina would become central players in lithium extraction and processing (Chile already produces both lithium carbonate and hydroxide), Brazil's well-established automotive sector could serve as a hub for manufacturing battery cells, LIBs, and EVs. In fact, China's BYD is moving ahead with plans to build a large-scale automotive production complex in Brazil, following the opening of its first factory for lithium iron phosphate in 2020.[72]

[70] UN Regional Commissions New York Office, *Catalyzing Transformative Actions and Integrated Approaches to Sustainable Development: Regional Experiences to Accelerate Progress on the SDGs*, October 2023, https://www.un.org/regionalcommissionsnyoffice/sites/www.un.org.regionalcommissionsnyoffice/files/2c2023_cn.pdf (accessed 25 June 2025).

[71] Mario Castillo, Ingrid Garcés and Rodrigo M. Furtado, *Perspectivas De Desarrollo De Las Cadenas De Valor Relacionadas Con El Litio En Chile Y América Del Sur,* Recursos Naturales y Desarrollo 223 (Santiago de Chile: Comisión Económica para América Latina y el Caribe [CEPAL], 2024), https://repositorio.cepal.org/server/api/core/bitstreams/7c55f70e-9192-4a8e-85c2-25e392bd349e/content (accessed 25 June 2025).

[72] BYD, *BYD Starts Operations at Its Lithium Iron Phosphate Battery Factory in Brazil*, 09.01.2020, https://www.bydglobal.com/sites/Satellite?c=BydArticle&cid=1514437377227&d=Touch&pagename=BYD_

Still, despite the clear benefits of closer regional collaboration, progress towards a coordinated industrial strategy has been limited. One of the main obstacles lies in the institutional and policy differences between countries in the Lithium Triangle: Argentina has taken a liberal, investment-driven approach; Bolivia continues to favour a state-controlled model; and Chile follows a mixed strategy that combines state oversight with growing incentives for private investors.

Broader regional integration also faces structural challenges. Intra-regional trade in South America remains weak, accounting for only about 15% of total trade, and thus well below North America's 38% and the EU's 55%.[73] External trade agreements are similarly fragmented: MERCOSUR recently concluded an FTA with the EU, while Chile and the EU have renewed their own association agreement, which now includes a chapter on critical raw materials. Chile, Peru, and Mexico also hold FTAs with the US, allowing their raw materials to benefit from incentives introduced under the IRA (although the future of those subsidies has now become uncertain under the current Trump administration).

In short, while national priorities continue to outweigh regional initiatives, there is clear momentum for technical cooperation, particularly among the lithium-rich countries of Latin America.

Recommendations

- **Synergies through regional collaboration:** As mineral-rich countries aim not only to ramp up production but also to generate local value and foster economic diversification, they often struggle with competitiveness, access to know-how, infrastructure development, and broader macroeconomic challenges. Stronger regional integration of supply chains could help address these issues by leveraging each country's comparative advantages, enabling scalability, and promoting regional trade, technical cooperation, capacity-building, and technology transfer.
- **Incorporate bi-regional cooperation into the EU's external raw materials strategy:** Beyond bilateral partnerships, the EU should give greater weight to bi-regional formats in its external raw materials strategy. Just as it has shifted critical raw materials policy from the national to the EU level to avoid fragmentation and inefficiency, it should apply the same logic to its external cooperation. For certain supply chains, supporting regional cooperation might be more beneficial to all countries as it creates synergies. A stronger emphasis on regional development could also reduce competition among resource-rich countries, which is only intensifying amid global demand and geopolitical rivalries, and counter the perception that bilateral deals are more advantageous than working with regional partners.
- **Support regional initiatives and institutions directly**: In Africa, support for the AMDC's continental work—and regionally through the SADC (e.g. support development of the SADC Industrial Strategy, especially around the battery value chain)—currently offers the most structured opportunities for engagement, but the EU should also engage the AfCFTA secretariat in the minerals sector. In Latin America,

EN%2FBydArticle%2FBYD_ENCommon%2FArticleDetails (accessed 14 July 2025); Leonardo Lara, "BYD Picks Brazil for Its First EV Hub Outside Asia", *Mining.com; Bloomberg News*, 4 July 2023, https://www.mining.com/web/byd-picks-brazil-for-its-first-ev-hub-outside-asia/ (accessed 14 July 2025).

[73] Shannon K. O'Neill, "Why Latin America Lost at Globalization—And How It Can Win Now: A Case for Greater Intraregional Trade in Today's Changing World", *Americas Quarterly*, 26 July 2022, https://www.americasquarterly.org/article/why-latin-america-lost-at-globalization-and-how-it-can-win-now/ (accessed 25 June 2025).

regional cooperation on raw materials is less developed, but EU support for institutions such as ECLAC, which promote regional strategies, should continue.

3 EU's External Raw Materials Strategy: Strategic Partnerships

The EU's Critical Raw Materials Act, adopted in May 2024, sets out a dual strategy: boosting domestic capacity and diversifying international supply chains to strengthen Europe's industrial base and safeguard critical inputs for the green, digital, and defence sectors. The European Commission has taken the lead in coordinating efforts within the EU and vis-à-vis external partners.[74] At the centre of the EU's diversification strategy lies a growing network of bilateral partnerships with resource-rich third countries.

This section examines how strategic partnerships are being implemented in practice, focusing on the cases of Zambia and Chile. Additional countries, including South Africa, Serbia, and Canada, were also analysed as part of the SWP projects and serve as additional reference points.[75]

International partnerships are central to the EU's external raw materials strategy. Zambia and Chile exemplify the diversity of partner contexts—differing regionally, economically, and in their engagement with the EU. Chile, a high-income country in Latin America, has long-standing institutional ties and deeper supply chain integration. Zambia, by contrast, is a developing country in Africa with less formalised cooperation and weaker global market linkages. This contrast allows for a nuanced comparison of implementation dynamics and shared structural challenges. Despite early political momentum, progress remains uneven due to limited industrial engagement, fragmented EU coordination, and a lack of dedicated financing.

These shortcomings are all the more relevant in the context of intensifying geopolitical competition. The CRMA's external dimension is unfolding in a dynamic global environment in which bilateral partnerships are not only technical but increasingly political. Whereas the EU is refining its engagement, actors such as China and the Gulf States have already been moving quickly to secure long-term, strategically advantageous agreements with resource-rich countries.

To remain credible, the EU must shift from declarations to delivery. Strategic partnerships require tangible investment, active private-sector participation, and targeted

[74] Meike Schulze, *Security of Supply in Times of Geo-Economic Fragmentation: Enhancing the External Dimension of the EU's Raw Materials Policy,* SWP Comment 2024/C 15 (Berlin: Stiftung Wissenschaft und Politik [SWP], April 2024), doi: 10.18449/2024C15.

[75] Schulze, "The Strategic Raw Material Partnership between the EU and Zambia" (see note 7); Carry, "Raw Materials Partner Chile: More Than Just a Supplier" (see note 6); Langalanga, Müller and Joseph, "South Africa and the EU Should Collaborate to Build Strategic Interdependence in Critical Minerals" (see note 64); Melanie Müller, Lea M. Strack and Marina Vulović, *The EU's Raw Materials Diplomacy: Serbia as a Test Case: The Rule of Law and Sustainability as Benchmarks for Europe's Raw Materials Cooperation,* SWP Comment 2025/C 10 (Berlin: Stiftung Wissenschaft und Politik [SWP], June 2025), doi: 10.18449/2025C10; Carry, "Critical Raw Materials Partner Canada: An (almost) Perfect Match" (see note 37).

financing. Strengthening coordination and embracing a coherent "Team Europe" approach will be essential in order to move from ambition to impact—and to ensure Europe helps shape, rather than merely react to, the future of global raw material supply chains.

Strategic Partnerships in the EU CRMA

The introduction of the CRMA marked a significant step towards a more cohesive raw materials policy within the EU. The directive aims to ensure that European industries have a resilient and sustainable supply of critical raw materials.[76] To achieve this goal, the CRMA comprises two pillars. Firstly, the expansion of intra-European mining, processing, and recycling of critical minerals; and secondly, the expansion of strategic partnerships on raw materials with resource-rich third countries.

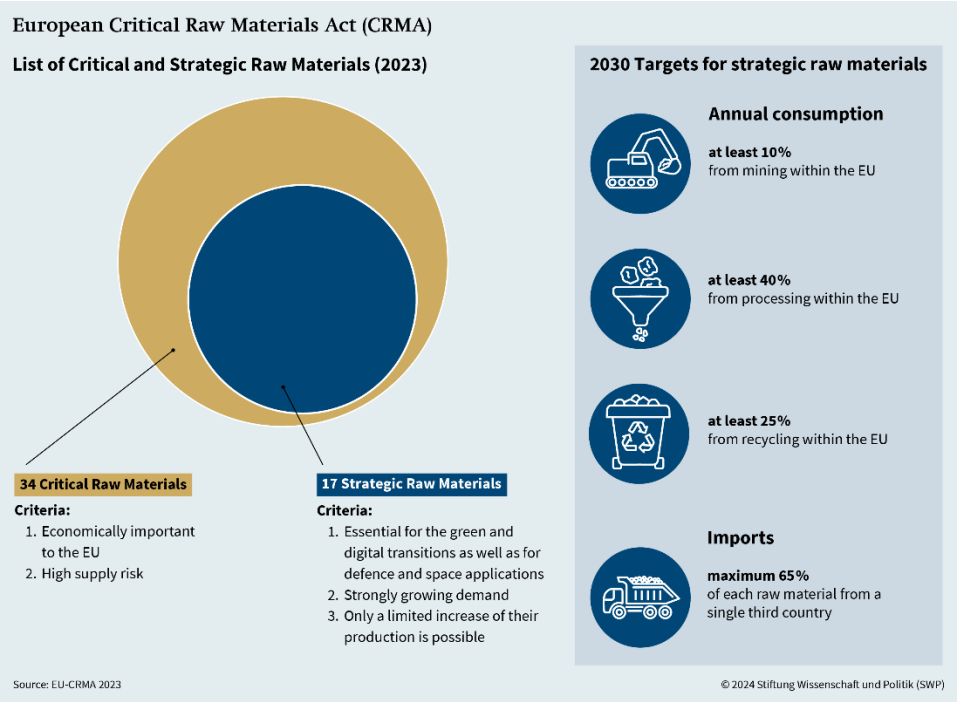
The CRMA's benchmarks are based on a list of critical raw materials, which was first introduced in 2011 and is updated every three years. In 2024, the list comprised 34 raw materials. Of these 34 raw materials identified as "critical", 17 are additionally classified as "strategic" based on their significance to the EU's industry and their high supply risk. The CRMA has established benchmarks for these strategic materials to be achieved by 2030. These targets include the expansion of the EU's capacity to mine, process, and recycle these strategic materials, and secondly, the diversification of non-European suppliers (see Figure 3).

To support the implementation of the CRMA, the EU established the European Critical Raw Materials Board (CRM Board), comprising representatives from the European Commission and member states, and tasked with overseeing and coordinating the implementation of the CRMA. A crucial tool for European action is the promotion of "strategic projects", which will be selected by the CRM Board and shall benefit from political support, including the identification of financing opportunities (see subsection "The Role of Public Finance in the EU's Delivery Gap"). Following the first call for applications, which closed in August 2024 and generated 170 submissions globally, the Commission published an initial list of 47 projects within the EU that were granted strategic status in March 2025. These projects cover 14 of the 17 strategic raw materials, with a central focus on those used in battery production.

The second key instrument of the CRMA is the EU's strategic partnerships on raw materials. Recognising that full self-sufficiency is neither feasible—due to geological constraints—nor timely, external supply diversification is essential. These partnerships are designed to foster new mineral projects and open trade to build secure, resilient supply chains. Between 2021 and 2025, the EU has entered into 14 such bilateral partnerships with resource-rich third countries (see Appendix). These partnerships are formalised through MoUs that are founded on a mutual interest to cooperate along raw material supply chains. The implementation is to be based on jointly developed roadmaps, aiming to establish "win-win situations" for all parties involved. In June 2025, the European Commission published an updated list of its selected strategic projects, which currently includes only 13 projects outside of the EU—nine of them located in official partner countries. Additional initiatives are planned in South Africa, Malawi, Madagascar, Brazil, the United Kingdom (UK), and New Caledonia.

[76] Schulze, "Security of Supply in Times of Geo-economic Fragmentation" (see note 74).

Figure 3: Key Elements of the EU CRMA



Source: Meike Schulze, *Security of Supply in Times of Geo-Economic Fragmentation: Enhancing the External Dimension of the EU's Raw Materials Policy,* SWP Comment 2024/C 15 (Berlin: Stiftung Wissenschaft und Politik [SWP], April 2024) .

Two broader domains should guide the further development of the EU's external raw materials strategy: First, the political rationale behind partner country selection. Although most partnerships involve democratically leaning countries, the underlying selection criteria—and more importantly, the EU's approach to navigating national and regional conflicts—remain vague. The second domain is the strategic role of partnerships within the EU's broader raw materials agenda. Although the number of partnerships is growing, particularly with developing and emerging economies in the Global South, their place within the evolving EU strategy is unclear. As the EU increasingly prioritises intra-European capacities and economic security, it remains ambiguous as to how these international partnerships are aligned with this shift. Despite the rhetoric about avoiding trade-offs, strategic tensions are visible.

Partner Diversity and Political Risks

The EU's current portfolio of bilateral raw materials partnerships is marked by considerable geographical, political, and economic diversity. In practice, the EU appears to prioritise countries with substantial strategic mineral reserves with the potential to host strategic projects. Many of these partnerships build on existing economic and diplomatic ties, suggesting that they were often formed in response to timely political openings rather than as part of a coherent, long-term strategy. Additionally, there are indications that the EU may have underestimated the political dimension of these partnerships, particularly as geopolitical tensions around access to critical raw materials continue to intensify. This oversight could become more consequential as global competition over raw materials deepens. Two broad categories of partner countries can be distinguished.

The first group consists of the EU's "allies", including industrialised mining nations such as Canada, Australia, and Norway, as well as EU-friendly emerging economies such as Chile and Argentina as well as Ukraine—the EU's first strategic raw materials partner. These countries represent relatively "low-hanging fruit" for the EU, given their established trade relations, alignment with EU (geo)political interests, and generally favourable investment environments, partly due to comparatively high mining standards. However, recent developments, such as the political shift in Argentina, illustrate that even among close allies, stable and dependable cooperation cannot be taken for granted. Ukraine further illustrates how quickly political conditions can change. Amid the ongoing conflict, Ukraine's minerals sector has attracted increased attention from the US, which is currently negotiating a "minerals for security" deal, which might also negatively the raw materials partnership with the EU. Nevertheless, serious doubts remain regarding the feasibility of launching or safely operating new mining projects under current conditions. The EU's 2025 designation of a strategic project in Ukraine may therefore be understood as a symbolic act of political support as much as a concrete industrial commitment.

The second, even more diverse group consists of mineral-rich countries such as Uzbekistan, Kazakhstan, Rwanda, the DRC, and the most recent CRM partner, Serbia. These countries are difficult to categorise uniformly: Whereas some exhibit democratic tendencies, others lean more towards authoritarian governance. What they share, however, are comparatively weaker political and economic ties with the EU, alongside structural challenges that diminish their investment attractiveness and complicate responsible mining practises.

Political dynamics also affect these partnerships. Rwanda's role in fuelling conflict in the DRC—particularly its links to the minerals sector—has raised doubts about the viability of continued EU cooperation and prompted the European Parliament to call for a suspension of the partnership with Ruanda. Similarly, ongoing and severe domestic protests in Serbia highlight how an authoritarian government, fragile rule-of-law structures, and the lack of implementation of environmental standards can lead to the public questioning—and ultimately undermining—the legitimacy of EU engagement in the raw materials sector in third countries.[77] Nevertheless, the European Commission has opted to maintain its support for the controversial Jadar project in Serbia. It was disclosed in June that the project is included among the EU's strategic projects in third countries.

Against this backdrop, the EU must make a special effort to shape and implement strategic partnerships with these countries. This requires not only stronger and more sustained political engagement, but also clearly defined red lines. At the same time, economic development and value addition is a key priority for these partner countries. EU principles such as fair deals and win-win cooperation are especially relevant here.

From Commitments to Delivery: The Implementation Challenge

The EU must now demonstrate that it can translate its bilateral raw materials partnerships into meaningful action. Rather than further expanding the partnership network, the priority should be on ensuring the effective implementation of existing ones. Failure to deliver on the ambitious goals set out in its MoUs, particularly with resource-rich countries in the Global South, risks damaging the EU's credibility and weakening its position in global raw materials diplomacy. Instead of pursuing ever more partnerships, greater focus is needed on responding to political dynamics (see subsection "Partner Diversity and Political Risks"), strengthening implementation and EU institutional

[77] Müller, Strack and Vulović, "The EU's Raw Materials Diplomacy: Serbia as a Test Case" (see note 75).

capacity, prioritising strategic projects, and addressing partner countries' concrete demands.

There are important lessons to be drawn from earlier partnership models. Germany's raw materials agreements with Mongolia (2011), Kazakhstan (2012), Chile (2013), and Peru (2014) serve as a cautionary tale. These partnerships largely failed to deliver, mostly due to declining interest from German companies amid falling commodity prices and a relaxed market environment in the 2010s, insufficient policy flexibility to adapt to varying partner contexts, inappropriate financing mechanisms, and a lack of German companies willing or capable of investing and operating in the upstream supply chain.[78] The EU should avoid repeating these mistakes. Particular attention must be paid to EU lead firms: Although the private sector now appears under greater pressure to diversify and secure supply chains—driven by rising demand, intensifying competition over access, and supply chain disruptions—early signs suggest that business transformation and long-term commitments to CRM partnership frameworks still fall short of the level and reliability often assumed in EU policy discourse.

The following discussion of core implementation challenges draws on a preliminary assessment of the EU's raw materials partnerships with Zambia and Chile (see summary in Box 2).[79] Despite marked differences in their economic structures, governance systems, and degrees of integration with the EU, both countries highlight key implementation challenges that extend beyond their individual contexts. Notably, like most mineral-rich countries in the Global South, both place strong emphasis on domestic value creation (as outlined in Section 1) and participate in regional initiatives aimed at strengthening cooperation and capacity in the minerals sector (see Section 2).

In both cases, foundational steps towards partnership implementation have been taken—notably through the signing of MoUs in 2023 and then during the development of joint implementation roadmaps, finalised in 2024. However, several challenges must be addressed to ensure that these partnerships evolve in a strategic manner.

Firstly, the EU's efforts are unfolding in an increasingly competitive geopolitical context. Alongside established actors, particularly China, the US, and G7 members—which have all intensified their engagement in the raw materials space—a range of newer entrants, including India, Turkey, and especially the Gulf States,[80] are stepping up their resource diplomacy, particularly in the Global South.[81] Whether and how these emerging

[78] Lukas Rüttinger, Fiona Schüler, Christine Scholl and Anna Bach, *Die Deutschen Rohstoffpartnerschaften - Analyse Der Umsetzung Und Ausblick: Auftragsstudie,* RohPolPress-Kurzanalyse 6 (Berlin: adelphi, Dezember 2016), https://bit.ly/3AF7Jvp (accessed 15 August 2024); Bundesrechnungshof, *Abschließende Mitteilung an Das Bundesministerium Für Wirtschaft Und Energie Über Die Prüfung Rohstoffpartnerschaften: Kapitel 6092 Titel 687 02* (Bonn: 1 October 2021), https://www.bundesrechnungshof.de/SharedDocs/Downloads/DE/Berichte/2022/rohstoffpartnerschaften-volltext.pdf?__blob=publicationFile&v=2 (accessed 14 July 2025).S. 10; as well as personal Interviews with government and industry representatives, 2024 .

[79] The two case studies are based on fieldwork conducted by the authors. Between October 2024 and February 2025, more than 60 interviews were carried out with stakeholders from the EU Commission, government officials in the partner countries, industry representatives, civil society, and researchers. The field work took place as part of SWP's research projects "International Raw Material Cooperation" and "Research Network Sustainable Global Supply Chains". Selected findings have been published in the following SWP Comments: Schulze, "The Strategic Raw Material Partnership between the EU and Zambia" (see note 7); Carry, "Raw Materials Partner Chile: More Than Just a Supplier" (see note 6).

[80] Meike Schulze and Mark Schrolle, *Saudi Arabia Strives to Become Major Player in Mineral Supply Chains: A Questionable Pillar of Europe's Diversification Strategy,* SWP Comment 2021/C 52 (Berlin: Stiftung Wissenschaft und Politik [SWP], November 2024), doi: 10.18449/2024C52.

[81] Susanna Fitzgerald, Phesheya Nxumalo and Thomas Scurfield, *Deals Without Details: Exploring State-State Mining Partnerships and Their Implications*, 6 February 2024, https://resourcegovernance.org/articles/deals-without-details-exploring-state-state-mining-partnerships-and-their-implications (accessed 7 July 2025); Pia

partnerships will ultimately materialise remains uncertain. However, it is clear that, unlike the EU, actors such as China and the Gulf States frequently act with greater speed and flexibility, enabling them to align their investments more closely with partner governments' priorities, especially in industrial and infrastructure projects. Moreover, outside of the G7 frameworks (see Section 4), the EU appears relatively isolated, with no clear engagement structure having been established with other emerging players in the raw materials sphere, even though there are some opportunities for cooperation with these actors.[82]

Secondly, although the EU has been effective in projecting its ambitions through the signing of MoUs and promoting "win-win" partnerships, its visibility and impact in partner countries remain uneven. One key reason is the complexity of the EU's institutional setup, which can slow decision-making and complicate efforts to ensure coherent and timely engagement. This is evident in the raw materials sector, where a unified Team Europe approach has yet to take shape. In both Chile and Zambia, the member states that are present in the country vary significantly in terms of their levels of engagement, and their contributions differ accordingly. In Chile, for instance, some countries have already established a solid base for private-sector engagement in the raw materials industry (Germany, in particular, stands out through its Competence Center Mining and Raw Materials), while others remain far less engaged. This fragmentation among member states is compounded by internal coordination challenges within the EU. The involvement of multiple actors across various Commission Directorates-General (DGs)—DG GROW, which leads negotiations on critical raw materials; DG INTPA, which is responsible for the development cooperation and the Global Gateway Initiative; and other DGs such as Trade and Energy—makes internal processes cumbersome. In Zambia, this leads to lengthy coordination cycles, partly overlapping objectives, and ultimately weakened overall impact.[83] More broadly, the growing number of EU frameworks—from Just Energy Transition Partnerships (JETPs) and Team Europe Initiatives (TEIs) to Global Gateway projects, bilateral energy and raw materials partnerships, and the more recent Clean Trade and Investment Partnership (CTIPs) announced with South Africa—reflects strong political ambition. Such cross-sectoral efforts, while valuable, can increase complexity and complicate implementation—as illustrated by the EU's engagement in Chile. While the creation of a TEI on green hydrogen aligns with efforts to build intersectoral linkages, it also introduces another layer of partnership, further complicating coherent implementation and the development of an integrated strategy.[84] Encouragingly, the partnership with South Africa provides a constructive example. Here, the EU is currently aligning its resource cooperation more cohesively and has agreed to strengthen cooperation during the EU-South Africa summit. This could serve as an example for other possible cooperation ventures.[85]

Beuter et al., *Mapping Africa's Green Mineral Partnerships*, 20 January 2025, https://afripoli.org/mapping-africas-green-mineral-partnerships (accessed 7 July 2025).

[82] Carry, "Raw Materials Partner Chile: More Than Just a Supplier" (see note 6); Schulze, "The Strategic Raw Material Partnership between the EU and Zambia" (see note 7); Barbara Lippert and Stefan Mair (Eds.), *Neue Verhältnisse – Schwierige Beziehungen: Europa – USA – »Globaler Süden«,* SWP-Studie 2024/S 24 (Berlin: Stiftung Wissenschaft und Politik [SWP], December 2024), doi: 10.18449/2024S24.

[83] Schulze, "The Strategic Raw Material Partnership between the EU and Zambia" (see note 7); Poorva Karkare*, The EU's Partnerships Around Critical Raw Materials: Do Its Ambitions Match Reality?*, 26 March 2025, https://ecdpm.org/work/eus-partnerships-around-critical-raw-materials-do-its-ambitions-match-reality%20 (accessed 7 July 2025).

[84] Carry, "Raw Materials Partner Chile: More Than Just a Supplier" (see note 6).

[85] Langalanga, Müller and Joseph, "South Africa and the EU Should Collaborate to Build Strategic Interdependence in Critical Minerals" (see note 64).

Box 2: EU Strategic Raw Materials Partnerships with Zambia and Chile

Key Objectives

- Ensure resilient access to critical raw materials and diversify EU supply chains
- Combine EU supply security with local value addition in partner country (*"win-win" approach*)

Key Cooperation Areas

- Supply chain integration (e.g. joint ventures, industrial cooperation)
- Research and innovation
- Infrastructure development
- Sustainable and responsible sourcing
- Capacity-building and governance support

Institutional Set-up and Implementation Progress

- MoUs signed with both countries in 2023; Roadmaps finalised in 2024 (not public)
- Partnerships coordinated by EU Delegation in partner countries, DG INTPA (Zambia), DG GROW, and external consulting agency (Chile)
- Progress on **technical cooperation** (e.g. joint research and ESG projects, geological surveys) and increase in **political and business-to-business engagement** (e.g. delegations, forums)
- Use of development finance (European Fund for Sustainable Development Plus (EFSD+) and member states funds); alignment with *Global Gateway Initiative*

Challenges and Gaps

- ***Institutional:*** Fragmented EU coordination rather than Team Europe approach
    - **Zambia:** Not all local embassies and programmes are involved; no joint financing commitments for key infrastructure
    - **Chile:** Multitude of sectoral partnerships and additional raw materials partnerships on bilateral member state level
- ***Value Chain and Investment:*** EU support is only partly aligned with partner countries' core interests; lack of industrial projects: "strategic partnerships" and "strategic project" labels (CRMA) have limited reach; high risk perception deters private capital
- ***Technical Cooperation*****:** Weak implementation support for industrial strategies aimed at promoting local processing and regional manufacturing cooperation
- ***Geopolitical Dynamics:*** EU in competition with other actors that are already established in the mining sector or moving fast to gain entry

Thirdly, although the promotion and implementation of high environmental, social, and governance (ESG) standards should be central to the EU's partnerships on raw materials, this approach must be carefully framed and adapted to the local contexts. In countries such as Chile, there is already a high level of awareness around ESG issues, supported by robust national sustainability frameworks—often formulated with the help of development cooperation agencies such as Germany's GIZ. In such contexts, what matters most is practical cooperation through joint R&D, technology transfer, and dialogue on the implementation of standards. Moreover, promoting ESG without parallel industrial engagement risks coming across as paternalistic—particularly when presented as a

unique selling point or positioned in contrast to China. This is especially relevant in Zambia, where local industrialisation and investment are key policy priorities. The Serbian case, in turn, demonstrates very clearly that concerns about the inadequate implementation of ESG standards can even jeopardise the implementation of raw materials partnerships, as fears about the negative impacts run high.[86] If the EU aims to position itself as a credible champion of sustainability, it must work together with local stakeholders to identify shared priorities and support their implementation. As the cases above demonstrate, this requires flexibility and context-specific approaches rather than uniform solutions.[87]

Fourth, and most importantly, industrial cooperation should be the cornerstone of the EU's raw materials partnerships—but it is not yet. One general challenge persists: the EU is currently prioritising domestic capacity-building and increasingly channels support towards projects within its own borders. At the same time, partner countries—this is evident in Chile and even more so in Zambia—often lack access to finance and concrete support for industrial project development. Although the European Commission maintains that there is space in the market for all actors, the reality is more complex. The sector remains highly competitive, especially in exploration, mine development, and refining—where China already holds considerable capacity.[88] Stronger EU support for genuine value chain integration is needed—especially with regard to the successful implementation of projects outside of the EU. In general, CRM partnerships continue to face major challenges in mobilising private-sector engagement. Private-sector investments, in upstream activities in particular, remain limited. This is partly due to the structural characteristics of European (and especially German) industry. Most companies are active in downstream segments of the value chain.[89] Larger firms remain hesitant to invest in upstream extraction and processing, while small and medium-sized enterprises often lack the resources and market leverage to diversify supply independently. Market-related constraints also play a role. Many EU firms seek relatively small volumes, making direct sourcing unattractive. There is also limited willingness to enter into long-term offtake agreements or commit to fixed pricing, with many still placing faith in future market stability.[90] More generally, unfamiliarity with the sector and insecurity regarding the role of certain raw materials in future technologies (especially different types of batteries) act as additional deterrents, alongside persistent shortcomings in incentives and public financing instruments (discussed in Section 4).

[86] Jovan Rajić, Tibor Moldvai, Mirko Popović, Johanna Sydow and Katja Giebel, *The Jadar Project, Serbia: History, Context and Concerns,* Briefing (Berlin: Heinrich-Böll-Stiftung, March 2025), https://www.boell.de/de/2025/03/04/jadar-project-serbia-history-context-and-concerns (accessed 30 June 2025); Heinrich Böll Stiftung, *Lithium Mining in Serbia: "An Open and Productive Debate Is Not Possible"*, 18 July 2024, https://www.boell.de/en/2024/07/18/lithium-mining-serbia-open-and-productive-debate-not-possible (accessed 30 June 2025); Müller, Strack and Vulović, "The EU's Raw Materials Diplomacy: Serbia as a Test Case" (see note 75).

[87] Carry, "Raw Materials Partner Chile: More Than Just a Supplier" (see note 6); Müller, Strack and Vulović, "The EU's Raw Materials Diplomacy: Serbia as a Test Case" (see note 75); Schulze, "The Strategic Raw Material Partnership between the EU and Zambia" (see note 7).

[88] Melanie Müller, Christina Saulich, Svenja Schöneich and Meike Schulze, *From Competition to a Sustainable Raw Materials Diplomacy,* SWP Research Paper 2023/RP 01 (Berlin: Stiftung Wissenschaft und Politik [SWP], February 2023), doi: 10.18449/2023RP01; IEA, "Global Critical Minerals Outlook 2025" (see note 1).

[89] Müller et al., "From Competition to a Sustainable Raw Materials Diplomacy" (see note 88).

[90] Private sector perspectives were gathered through interviews conducted during the case studies, as well as through background discussions held at various events and industry fairs in 2024 and 2025.

The Role of Public Finance in the EU's Delivery Gap

With the adoption of the CRMA, the EU has opted for a relatively restrained approach. Unlike countries such as Japan and South Korea, which operate state-backed agencies with dedicated investment mandates,[91] the EU relies largely on private-sector engagement, supported by regulation and political signalling. Strategic partnerships and selected strategic projects along the supply chain form key pillars of this strategy. However, both in the run-up to the legislation and during early implementation of the CRMA, a significant financing gap has become apparent and also concerns the delivery capacity in strategic partnerships.

Efforts to provide more financial support exist at both the EU and member state levels. However, a key obstacle remains: Private banks and institutional investors in the EU are generally hesitant to enter the mining sector, while many major financiers are based outside of Europe.[92] Yet, the need for long-term project finance is persistent, and financing remains constrained due to commodity price volatility, fluctuating demand forecasts for transition minerals, and the inherently high-risk nature of the sector. Although geopolitical developments and political pressure has prompted some public banks, such as the European Investment Bank (EIB), and parts of the European private sector to re-engage, there is little sign of a fundamental shift in financing patterns. The broader economic climate—marked by the increasing fragmentation of global trade frameworks and industrial policy interventions—will further weaken the willingness to commit to long-term investments, particularly abroad.

At the same time, countries such as the US and many EU member states are increasingly seeking to deter Chinese investment in their critical raw materials projects to reduce the influence of Chinese (often state-owned) companies. Although this reflects geopolitical concerns, it has contributed to a growing financing shortfall, particularly for junior mining companies, which depend on high-risk capital.[93] The resulting vacuum creates further opportunities for Chinese and other state-backed investors, notably from the Gulf, who continue to benefit from their ability to act swiftly and with strategic intent.

At the EU level, financing remains highly fragmented, with no clear connection between strategic projects and targeted funding channels. Instead of establishing a dedicated EU raw materials fund, the CRMA proposes to optimise existing public programmes and accelerate private investment through faster permitting and regulatory streamlining. A finance group under the CRM Board—comprising the EIB, national promotional banks, and private-sector actors—has been tasked with advancing this agenda.[94] However, the CRMA has been widely criticised for its limited financial ambition: No dedicated EU budget or funding instrument has been created. Recently, in March 2025 the EIB launched the Critical Raw Materials Strategic Initiative, which aims to provide up to €2 billion in financing for CRM investments this year. In addition, a CRM Task Force and a one-stop-

[91] Hyun Soo Cho and Jane Nakano, *The Minerals Security Partnership Under the South Korean Leadership*, January 28, 2025, https://www.csis.org/analysis/minerals-security-partnership-under-south-korean-leadership (accessed 25 June 2025); Margaret Armstrong, Rafael D'Arrigo, Carlos Petter and Alain Galli, "How Resource-Poor Countries in Asia Are Securing Stable Long-Term Reserves: Comparing Japan's and South Korea's Approaches", *Resources Policy* 47 (2016): 51–60.

[92] Manuel Rico, *Mining for Minerals Is Not a European Business*, 26 October 2023, https://www.investigate-europe.eu/posts/mining-minerals-not-european-business (accessed 7 July 2025); Violaine Faubert, Nathan Guesse and Julien Le Roux, *Capital in the Twenty-First Century: Who Owns the Capital of Firms Producing Critical Raw Materials?,* SUERF Policy Brief 944 (Vienna: August 2024), https://www.suerf.org/wp-content/uploads/2024/07/SUERF-Policy-Brief-944_Faubert-et-al.pdf#page=4 .

[93] Carry, "Critical Raw Materials Partner Canada: An (almost) Perfect Match" (see note 37).

[94] Schulze, "Security of Supply in Times of Geo-economic Fragmentation" (see note 74).

shop for advisory services are to be established to enhance technical capacity and promote cross-sector partnerships. Yet, the design and implementation modalities of these measures remain vague. A more concrete step was announced in June 2024, when the EU and the European Bank for Reconstruction and Development (EBRD) launched a joint equity facility under the InvestEU framework. This fund will provide up to €100 million in equity capital for strategic CRM projects. The fund is open to projects in EU and EBRD partner countries, including Serbia and Ukraine, and will focus particularly on exploration and junior mining companies.

Furthermore, the Global Gateway Initiative plays a critical role in the implementation of strategic partnerships—particularly in infrastructure and accompanying measures, which are especially relevant for developing countries. Coordinated by the European Commission's DG INTPA, the initiative relies heavily on the implementation capacities of the *EIB*. The largest financial instrument under this framework is the *EFSD+.*[95] Although the EFSD+ supports a broad range of development priorities, only a limited share of its funding is currently directed towards mining and infrastructure projects relevant to raw materials value chains.

The limited financial capacities at the EU level highlight the continued reliance on member state initiatives—not only for industrial projects, but also for accompanying measures and technical cooperation. Germany's Raw Materials Fund, operational since 2024, foresees €1 billion in investments by 2027, offering €50–150 million per project through the KfW as a minority stakeholder. Political and commercial risks can be covered by untied financial loan guarantees (UKF) issued by Euler Hermes. However, project implementation has been delayed due to pending reviews of corporate investment projects. France and Italy have also announced national raw materials investment vehicles. France launched a €500 million fund in late 2024, managed by the private firm AfriVia, with the goal of scaling it to €2 billion. Italy included raw materials as a target sector under its broader "Made in Italy" fund, promising around €1 billion in public investment. However, no formal progress has been reported on CRM project funding. The Netherlands announced a similar public–private investment initiative in March 2025. Yet, it remains unclear how well these national efforts are coordinated within the CRM Board or aligned with EU-wide strategic objectives.

Furthermore, from national initiatives to the EBRD and the EU's Global Gateway Initiative, there remains considerable reliance on public incentives and guarantees to mobilise private investments—the "crowding-in" effects.[96] However, the effectiveness of this approach remains limited in the raw materials sector and has a potentially negative outlook, given the persistent lack of engagement by institutional investors. This challenge is particularly acute in mineral-rich developing countries, which continue to face major barriers in accessing capital and attracting private-sector involvement (see discussion in Section 2).[97]

[95] Simone Tagliapietra, *The Global Gateway: An Overview* (Washington, DC: The Wilson Center, August 2022), https://www.wilsoncenter.org/sites/default/files/media/uploads/documents/Global%20Gateway%208-1-22.pdf (accessed 19 August 2024).

[96] Ibid.

[97] Francesco Findeisen and Yann Wernert, *Meeting the Costs of Resilience: The EU's Critical Raw Materials Strategy Must Go the Extra Kilometer,* Policy Brief (Berlin: Hertie School, Jacques Delors Centre, June 2023), https://www.delorscentre.eu/en/publications/eu-critical-raw-materials (accessed 14 August 2024); Arthur Leichthammer, *Mining for Tomorrow: The Strategic Importance of Critical Raw Materials for Europe's Industry,* Policy Position (Berlin: Hertie School, Jacques Delors Centre, May 2024), https://www.delorscentre.eu/fileadmin/2_Research/1_About_our_research/2_Research_centres/6_Jacques_Delors_Centre/Publications/20240521_Mining_for_Tomorrow_Policy_Position_Arthur_Leichthammer.pdf (accessed 14 August 2024).

Recommendations

- **Strategic partnerships must be more clearly embedded in the EU's broader raw materials and industrial policy.** To enhance effectiveness, partnerships should clarify their role as instruments for strategic resilience rather than engaging in mere diplomatic alignment; prioritise a limited number of high-impact regions and countries to ensure credible implementation; and align with partner country priorities by delivering on industrial development, value addition, and strategic engagement.
- **Prioritise the implementation of a unified Team Europe approach.** Effective ground-level delivery requires a coherent EU approach that strengthens inter-directorate coordination (especially between DG GROW and DG INTPA), simplifies budget access, and establishes dedicated raw materials leadership, while ensuring member states such as France and Germany coordinate their national instruments under a unified framework led by the European External Action Service (EEAS) in partner countries.
- **Ensure technical cooperation remains a core delivery element of partnerships, with tailored support through joint EU–partner country roadmaps**: Key priorities include providing financing and technical advisory services for mining and industrial policy alongside transparent infrastructure commitments in transport and energy, while fostering good governance and sustainability through international standards and respectful dialogue that honours local priorities.
- **Actively support industrial projects and supply chain development in partner countries:** Key measures include raising European company awareness of investment opportunities and strategic projects while addressing engagement barriers, expanding the EIT Raw Materials network globally to facilitate cross-border consortia, and implementing stronger incentives such as local content requirements and regulatory diversification rules to drive private-sector participation.
- **Strengthen the financial architecture to support raw materials partnerships**: The EU should make use of a broader repertoire of carrots (e.g. de-risking tools, long-term offtake structures) and sticks (e.g. diversification requirements) to align private-sector engagement with strategic supply goals. Key financial measures include creating a dedicated EIB-managed EU fund for strategic industrial projects with tailored developing-country approaches, significantly boosting Global Gateway Initiative funding beyond 2028 for sustained autonomous action, and securing long-term financing for technical measures despite declining member state development budgets. At the same time, the EU must send a stronger signal to companies and business associations to equally take up responsibility for devising a long-term strategy to secure their supply of critical minerals.
- **Develop a strategic approach to engagement with other geopolitical actors in partner countries that reflect local and global realities**. Strengthen coordination with like-minded partners such as the G7, but also pursue pragmatic channels of engagement with influential players such as China and the Gulf States.

4 Navigating a Fragmented Global Order

With the CRMA, the EU aims to diversify its mineral supply chains and strengthen its position in the global raw materials sector. In doing so, it is operating in an increasingly contested geopolitical landscape, where international cooperation is becoming more fragmented and transactional. So far, the EU has placed strong emphasis on bilateral partnerships to advance its strategic objectives, as outlined in the previous section. At the same time, it must recognise the need to engage beyond the existing formal bilateral partnerships by mainstreaming minerals cooperation in other bilateral partnerships and engage multilaterally to mobilise and de-risk investments along mineral supply chains. Multilateral fora also remain key domains in terms of the governance of mineral supply chains—including trade relations, investment conditions, and international standard-setting. This section highlights opportunities for further European engagement in the minerals sector.

The re-election of Trump presents a big challenge for the European Commission. Whereas the Biden administration supported multilateral initiatives for raw materials partnerships, particularly within the G7 and Minerals Security Partnership (MSP) framework, the Trump administration prioritises unilateral action and bilateral deals. The EU can therefore no longer assume US support for joint efforts to diversify supply chains, maintain open markets, and promote cooperative governance. This shift requires the EU to adjust its approach—in responding to the new US course, in continuing multilateral cooperation with partners still committed to such frameworks, but also in cooperation with resource-rich partners, especially in the so-called Global South.

The shifting geopolitical landscape raises important questions about how established and emerging powers, especially China and the Gulf States, are leveraging the current moment to expand their influence globally. The EU must therefore consider how to engage with these actors while acknowledging that they may act as both counterparts and competitors in shaping global minerals governance, but in certain areas potentially also as cooperation partners.

The EU's strategy towards key actors in the critical raw materials landscape—allies and competitors alike—must address two major issues. First, the financing of diversification efforts remains one of the main challenges in implementing the EU's critical minerals agenda. A central pillar of the EU's approach should therefore be determining to what extent, and in what ways, joint economic and financial cooperation is feasible. Second, the EU must consider how multilateral engagement on critical raw materials can be used to set international standards that reflect its outspoken commitment to sustainability and due diligence.

The Future of US and G7 Cooperation: Forming a "Coalition of the Willing" in the Minerals Sector

Until February 2025, one of the central pillars of the EU's strategy on critical raw materials was coordination with "like-minded states". In 2024, the EU launched its own Critical Raw Materials Club to bring together both consumer and producer countries to foster cooperation in raw material supply chains. Nevertheless, much of the EU's multilateral engagement has taken place as part of the Minerals Security Partnership, initiated by the Biden administration in 2022.

The MSP has since evolved into a key example of the "friend-shoring" strategy. It was established as a coordination platform for the G7 and other "like-minded" industrialised nations, with the shared objective of enhancing the resilience of mineral supply chains. This includes a clear geo-economic ambition: to develop alternatives to China's dominant role in the processing of critical minerals and, to a lesser extent, to reduce dependence on Russian mineral exports.[98] The partnership currently includes 15 members: Australia, Canada, Estonia, Finland, France, Germany, India, Italy, Japan, Norway, South Korea, Sweden, the UK, the US, and the EU. Activities pertaining to the MSP focus on four priorities: (1) diversifying and stabilising mineral supply chains, (2) boosting investment, and (3) increasing recycling.[99]

In 2024, the MSP Forum was created as an expanded platform to the core membership. Apart from MSP members, it also includes an additional 15 countries with key mineral reserves, primarily from the Global South, among them Argentina, Chile, Peru, the DRC, Zambia, as well as Greenland and Serbia. Initially, the forum aimed to support the policy coordination and good governance in the minerals sector. Over time, however, the priority shifted towards promoting projects and facilitating investment. By September 2024, the MSP was officially supporting 32 projects: 13 in Africa, 8 in the Americas, 6 in the Asia-Pacific, and 5 in Europe.[100] Although the partnership aims to facilitate projects along the entire value chain, the majority of the projects are concentrated in upstream and midstream processing, and support for value addition is still limited.[101] Moreover, general challenges persist in attracting private-sector interest and engagement, as well as in mobilising sufficient financial support to compete with more integrated approaches, as championed by China, particularly in the Global South.[102]

Growing (geo-) political tensions—not just with China but also among the G7 countries and the US—have created an increasingly uncertain future for the MSP. The outcome of the G7 summit in Canada in June 2025 has reaffirmed a shared commitment to mineral engagement through a G7 Critical Minerals Action Plan. The plan's core mission centres

[98] Günther Maihold, *A New Geopolitics of Supply Chains: The Rise of Friend-Shoring,* SWP Comment 2022/C 45 (Berlin: Stiftung Wissenschaft und Politik [SWP], August 2022), doi: 10.18449/2022C45; Vlado Vivoda, "Friend-Shoring and Critical Minerals: Exploring the Role of the Minerals Security Partnership", *Energy Research & Social Science* 100 (2023): 103085.

[99] Jane Nakano, *Chapter 9: An Evaluation of the Minerals Security Partnership,* in *Critical Minerals and the Future of the U.S. Economy*, ed. Gracelin Baskaran and Duncan Wood, 65–70., https://csis-website-prod.s3.amazonaws.com/s3fs-public/2025-02/250210_Baskaran_Critical_Minerals.pdf?VersionId=Tfu2TnNrQGlN7ol8HSCakMUT8HTwYukd (accessed 30 June 2025).

[100] Hyun Soo Cho and Jane Nakano, "The Minerals Security Partnership Under the South Korean Leadership".

[101] Scurfield, Salomon and Olan'g, "Six Keys to Unlocking Equitable Value Addition in Mining" (see note 3).

[102] Gracelin Baskaran and Meredith Schwartz, *G7 Cooperation to De-Risk Minerals Investments in the Global South*, 7 May 2025, https://www.csis.org/analysis/g7-cooperation-de-risk-minerals-investments-global-south (accessed 25 June 2025).

around creating standards-based markets, mobilising capital, and promoting innovation. However, despite this renewed unity, three areas of potential division are emerging:

- *Shifting strategic purpose:* The new G7 statement acknowledges a declining emphasis on green technologies. The US, in particular, is redirecting focus towards the defence and security industries. This shift risks creating conflicting objectives within the MSP membership, but also with third countries active in the MSP Forum.
- *Diverging economic approaches:* The partnership raises the question of active state involvement in markets and value chains. Countries such as South Korea and Japan have long pursued industrial policies through (semi-)state-owned raw materials agencies, whereas others—such as Australia—prefer open-market, private-sector-led approaches.
- *Uncertainty over US leadership:* Under President Trump, the US has made it clear that national economic security takes precedence, with a strong preference for bilateral deals in which critical minerals are used as bargaining chips in trade or security negotiations. This approach casts serious doubt on the future of US support for both the G7 Action Plan and the MSP. Trump's previous threats—economic and rhetorical—towards key allies such as Canada, Australia, and the EU have already raised questions about the long-term viability of these multilateral frameworks. Even now, there are signs of waning US engagement that suggest a more uncertain trajectory going forward.

Three scenarios seem plausible: The most optimistic—but least likely—is that the Trump administration stays actively engaged in multilateral efforts on critical minerals within the G7 and the MSP. However, Trump's open disdain for multilateralism, his repeated economic provocations aimed at allies (particularly Canada and the EU), and his tendency to resist actions that appear to benefit the EU make sustained engagement an unlikely path. A more probable scenario is that the US takes a back seat, remaining formally within the MSP but offering little in terms of leadership or initiative. Although the partnership could continue under such conditions (with South Korea holding the MSP chair through the summer of 2025), its overall impact and cohesion would be diminished without active US involvement. The third—and not improbable scenario—is more disruptive: The US could move to actively undermine G7 coordination efforts on critical minerals if multilateral approaches are seen as clashing with US economic goals. In this case, Trump might use obstruction as a way to extract bilateral concessions or leverage his position in alternative deals negotiated outside of these multilateral forums. Such a turn would not only weaken the MSP, but it could also further strain transatlantic cooperation at a time when alignment is urgently needed.

What does this mean for finance and investment coordination?

In September 2024, the MSP launched its finance network, bringing together more than 30 institutions, including member states' export credit agencies and development banks. The initiative seeks to share information, support sustainable investments, and mobilise private capital for both existing and future MSP projects. These ambitions were recently reaffirmed in the G7 Action Plan, which emphasises ongoing collaboration in the financial domain. However, the, prospects for significantly strengthening this forum, or for eventually creating a jointly managed international raw materials fund, remain limited. The current approach continues to prioritise national-level activities and the use of financial instruments to secure domestic value creation, rather than directing substantial support to partner countries (see Section 3). This illustrates a growing emphasis on

national interests in the raw materials sector, often at the expense of cooperative approaches.

In addition, the shifting political priorities in the US further complicate the landscape. As a key financier, the US appears increasingly inclined to pursue bilateral agreements and is hesitant to fund international projects that do not clearly serve its strategic interests. This casts doubt on its long-term commitment to multilateral initiatives. As a key financier, the US appears increasingly inclined to pursue bilateral deals and reluctant to fund foreign projects that do not clearly align with its strategic interests. This casts doubt on its long-term commitment to multilateral initiatives. This dynamic also affects another financing effort: The G7's Partnership for Global Infrastructure and Investment, which launched in 2022 with the goal of mobilising $600 billion in 2027 across sectors such as transport, energy, and supply chain resilience, has shown some promise—for example, through the US-EU push for support for the Lobito Corridor, an infrastructure link between Zambia, the DRC, and Angola. Yet, the long-term reliability of such commitments remains uncertain.

How to move forward

This shifting geopolitical landscape underscores the urgent need for the German government and the EU to intensify cooperation with both consumer and producer countries that share a commitment to diversified supply chains, multilateralism, and sustainability. Within the G7, partners such as Canada[103] and Australia stand out. Their strong political and trade relations with Europe, shared geopolitical outlook, and deep expertise in mining and hosting major mining companies make them natural allies. The EU should therefore deepen its collaboration with these countries and explore options for continuing MSP-related efforts without US participation. Instead of creating yet another multilateral initiative, such as a rebranded MSP or a new EU Raw Materials Club, the EU could focus on forming a "coalition of the willing" within existing formats, such as the G7 and the MSP. Within these formats, the EU should also examine possibilities for collaboration with other resource-rich Global South countries, whether through trilateral cooperation or broader arrangements.

One critical pillar of such a coalition should be a focus on technical cooperation. One important aspect is streamlining permitting processes. Faster and more predictable permitting can offer a significant competitive advantage by attracting private capital. However, this drive for efficiency must not compromise environmental and social safeguards. A "race to the bottom" on sustainability would undermine long-term resilience, which is not viable in many partner countries where domestic regulations and protections for indigenous and local communities are foundational (see also Section 3 and Conclusion). For the EU and Germany, maintaining high sustainability standards must remain non-negotiable. The focus should be on making these standards more implementable and transparent—not weakening them. This includes a commitment to horizontal and sector-specific corporate responsibility due diligence regimes to ensure high sustainability standards along the entire supply chain. While due diligence legislation is increasingly being challenged by certain actors within the EU,[104] other countries such as

[103] Carry, "Critical Raw Materials Partner Canada: An (almost) Perfect Match" (see note 37).

[104] Jack Schickler, "EU's Threatened Corporate Due-Diligence Rowback Raises 'Deep Concerns' from Left", *Euronews*, 17 December 2024, https://www.euronews.com/my-europe/2024/12/17/eus-threatened-corporate-due-diligence-rowback-raises-deep-concerns-from-left (accessed 9 July 2025); Inga Carry and Svenja Schöneich, *Beyond the First Tier: Implementing Due Diligence in Raw Material Supply Chains - Research Network Sustainable Global Supply Chains*, 03.03.3035, https://www.sustainablesupplychains.org/blog/beyond-the-first-tier-implementing-due-diligence-in-raw-material-supply-chains/ (accessed 7 August 2025).

Canada, Australia, and South Korea are making tangible progress towards more comprehensive regulatory regimes. Cooperation among other resource-rich countries wanting to join the "coalition of the willing" should therefore also include a joint commitment to the advancement of due diligence regimes, both to reinforce shared values and to contribute to a more level global playing field.

The second critical pillar should be on financial cooperation. Such a coalition should prioritise the development of joint financing and investment vehicles that are designed to de-risk critical mineral projects and improve competitiveness along the supply chain. It should also work to identify potential funding and collaboration opportunities in partner countries, especially in those of the Global South. The South African government, for example, has already established an exploration fund that aims to support junior mining companies, which could facilitate bilateral cooperation in this field. Beyond identifying new resource projects, the coalition should promote strategic vertical integration, ensuring that key segments of the supply chain—particularly processing and refining—are retained within partner countries. Simply extracting raw materials for export to China for processing only perpetuates existing dependencies. Future investments could be deliberately structured to align with broader economic and geopolitical objectives.

Multilateralism in Motion: The Global South's Agency and Strategic Diplomacy

Amid ongoing geopolitical realignments, it is more crucial than ever for the EU to clarify its position and develop a unified, forward-looking strategy towards the Global South. The bilateral raw materials partnerships outlined in Section 3 represent merely an initial step. The EU must not only deliver on these commitments credibly, but also broaden its toolkit—both by integrating raw materials partnerships into other forms of existing cooperation, and by adopting a more proactive stance as a raw materials partner in multilateral forums.

Looking ahead, the EU also has a strategic interest in deepening cooperation with countries that advocate for open and rules-based approaches to raw materials governance and whose national frameworks promote transparency, economic cooperation, and sustainability. These include countries such as South Africa, Brazil, Chile, Peru, Zambia, and Namibia, which combine resource endowments with a willingness to engage in international partnerships and uphold shared values in areas such as ESG standards and compliance with trade rules. Formal partnerships and MoUs are not always necessary, as they often involve lengthy political and bureaucratic processes while generating expectations. Instead, integrating raw materials components into existing bilateral partnerships can be more effective, for example by incorporating raw material elements into formats such as the JETPs, TEIs, and the more recent CTIPs. This approach allows EU offerings to be better aligned and customised to specific needs.

At the same time, the EU is right to advocate for open markets and to condemn market disruptions such as those caused by China. However, it should not dismiss engagement with countries that pursue protectionist strategies aimed at local value creation and development. This is the case in nearly all mineral-rich countries, albeit to varying degrees and through different instruments. One example is Indonesia, where export restrictions on unrefined nickel—intended to build domestic processing capacity—have prompted the EU to file a case at the WTO.[105]

[105] Inga Carry, Lea Strack and Melanie Müller, *Running the Race for Critical Minerals – Between Geopolitical Rivalries and Sustainable Development,* in *Sustainable global supply chains in times of geopolitical*

Crucially, the EU also needs to recognise that many resource-rich countries, especially in the Global South, do not necessarily share Europe's geopolitical priorities. Instead, they often seek to work with a range of partners, including both Western countries and actors such as China. Trying to force these countries to choose sides—or to distance themselves from partners who currently offer critical financing, technology, or infrastructure—is unlikely to succeed, particularly if the EU is not yet in a position to offer credible alternatives. If the EU wants to remain relevant and build genuine partnerships, European companies and institutions will need to step up: by providing technologies that help make mining more efficient and environmentally responsible; by offering training; and by investing in local capacity. Otherwise, it is only logical that countries will look elsewhere to develop capabilities, especially along the downstream supply chain.[106] As mineral-rich countries seek to strengthen local value addition, cooperation—especially on critical raw materials—remains essential. Disengagement would risk deepening existing divides and weakening Europe's ability to collaborate effectively in the minerals sector. Instead, the EU should assess—on a case-by-case basis—how it can offer mutually beneficial partnerships. A more flexible and differentiated approach would enable the EU to remain a relevant and constructive partner across a broader spectrum of political and economic contexts.

Strengthening cooperation in the mineral resources sector must go beyond bilateral engagements. Depending on the context, regional cooperation should also be given greater emphasis, for example by engaging AfCFTA in conversations on mineral supply chains. Crucially, new offerings must correspond with concrete engagement of European companies, and thus also serve the economic interests of partner states. It also requires proactive involvement in multilateral forums that shape global norms and investment flows in this space. The South African G20 presidency presents a timely opportunity for the EU to deepen its cooperation with other G20 members in the minerals sector. South Africa has named critical raw materials as one of the five priorities of its G20 agenda and is working to initiate joint efforts at the G20 level. This offers an opportunity to engage with resource-rich countries from different world regions and develop a framework for joint cooperation in the minerals sector. The South African government has repeatedly emphasised its commitment to global cooperation, free trade, and the implementation of ESG standards. The EU should support the efforts of the South African government at the G20 level.[107]

Apart from the G20, other multilateral forums that span rising geopolitical divides are essential—especially for international standard-setting and establishing ESG frameworks for raw materials projects and investments. The recently formed UN Panel on Critical Energy Transition Minerals, co-chaired by the South African ambassador and the EU Director-General for Energy, should become a key format for the EU's global dialogue. Forums such as the IGF, and the Extractives and Transparency Initiative (EITI) remain relevant for continued engagement and provide an opportunity to engage in standard-setting and implementation. Such formats could help mitigate the current trend towards bloc formation in mineral trade and governance.

Cooperation with developing and emerging economies requires more than political will—it depends on concrete financial support for governance reforms, technical

crises: Annual Report 2023, ed. Stiftung Wissenschaft und Politik (SWP), 58–87., https://www.sustainablesupplychains.org/wp-content/uploads/2024/01/SustainableGlobalSupplyChains-Report2023.pdf (accessed 30 June 2025).

[106] Carry, "Raw Materials Partner Chile: More Than Just a Supplier" (see note 6).

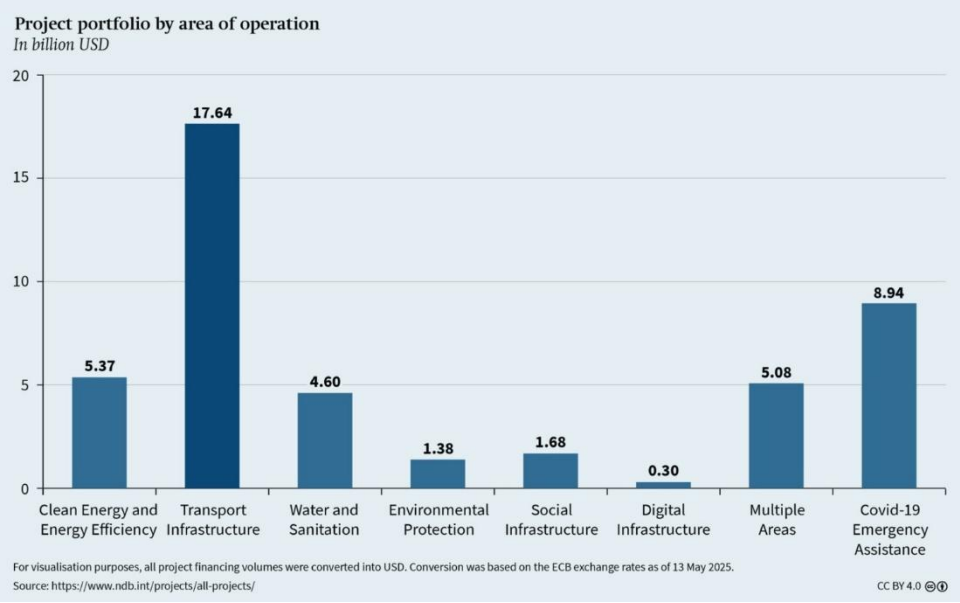
[107] Langalanga, Müller and Joseph, "South Africa and the EU Should Collaborate to Build Strategic Interdependence in Critical Minerals" (see note 64).

cooperation, and infrastructure. Although the MSP Finance Group and the G7's infrastructure initiative have been launched, they continue to focus on specific projects and largely reflect the interests of industrialised countries. What is needed is more flexible, locally anchored support—particularly through multilateral and regional development banks. The EU should identify all possible options.

The World Bank remains a key multilateral actor, especially as it has slowly begun to recognise the importance of industrial policy for boosting local value creation. A recent example is the launch of the Resilient and Inclusive Supply-chain Enhancement (RISE) Partnership in October 2023 under Japan's G7 presidency. Designed as a multi-donor platform, RISE aims to help developing countries strengthen their manufacturing base and better integrate into mineral value chains. However, there has been minimal initial participation in the fund. Despite renewed G7 commitments to provide stronger financial backing, tangible progress is lacking. Uncertainty also surrounds the future role of the World Bank, particularly given its dependence on US support—the institution's largest shareholder.

Regional development banks already play a critical role. Although they rarely fund mining projects directly, they are central to advancing regional industrial strategies. The Asian Development Bank and Inter-American Development Bank, for instance, are involved in large-scale energy and transport infrastructure. The AfDB is especially noteworthy—not only as a co-developer of the African Green Minerals Strategy, but also as a major financier, contributing to roughly a quarter of all infrastructure projects across the continent.

Figure 4: NDB projects: Project portfolio by area of cooperation



Source: Melanie Müller, "Between Global Geopolitics and National Interests: BRICS Cooperation in the Mineral Sector", *South African Journal of International Affairs* 32, 1-2 (2025): 215–241.

Within BRICS, raw materials cooperation remains limited. Although member states agreed to strengthen collaboration in the minerals sector at the 2023 summit in Kazan (Russia), the focus is primarily on enhanced cooperation between geological services with

the potential goal of establishing a joint "BRICS+ geological platform".[108] However, the BRICS New Development Bank has yet to finance any significant raw materials projects. Of the 149 projects published on the bank's website as of April 2025, none explicitly targets the raw materials sector. Financing very often focuses on infrastructure development, which may benefit the minerals sector but is not specifically designed for it. Only one project explicitly mentions contributions to the minerals sector.[109]

Moreover, most BRICS+ members—with the exceptions of Russia, Iran, and to some extent the United Arab Emirates (UAE)—are pursuing their own geopolitical and economic objectives. These primarily involve building partnerships with various actors in different world regions, rather than developing targeted raw materials cooperation within the BRICS format itself.[110] This also underlines the necessity for the EU to engage with actors in different world regions.

Engaging Complex Partners: China and the Gulf States

The EU's international raw materials strategy must also take into account its role and engagement with two key actors: China, a well-established player that retains a dominant and strategic position in these supply chains, and the Gulf States, particularly Saudi Arabia, which—alongside the EU—are emerging as potentially significant actors. Despite growing efforts by the EU and other G7 countries to diversify supply sources, China remains a key supplier of critical minerals in the short to medium term, as well as a central partner for many resource-rich countries in the Global South. At the same time, the Gulf States are increasingly pursuing strategic resource diplomacy and forging new partnerships and investment linkages across the minerals sector. This shifting landscape will affect both the EU's supply security as well as its cooperation models in third countries.

The EU-China relationship is increasingly shaped by geopolitical and geo-economic tensions, which are complicating opportunities for cooperation, particularly in the critical minerals sector. Although the EU had traditionally taken a more moderate diplomatic stance towards Beijing compared to the US, this position is now shifting. In response to growing concerns over market imbalances and unfair competition, the European Commission is now pursuing a more strategic "risk-management approach"[111] and has introduced more assertive trade protection measures. This change in posture has been reinforced as a consequence of China's expanding use of export restrictions on raw materials and strategic technologies. The most recent example is China's decision to implement export licensing requirements for seven REEs and related magnets, which have also led to severe consequences for European industries, particularly the automotive and electronics industries.[112] Although these measures are part of the broader geopolitical

[108] *Kazan Declaration: Strengthening Multilateralism for Just Global Development and Security* (23 October 2024), https://brics.br/pt-br/documentos/acervo-de-presidencias-anteriores/leaders-declarations/2024-xvi-brics-summit-kazan-declaration.pdf/@@download/file (accessed 30 June 2024).
[109] Melanie Müller, "Between Global Geopolitics and National Interests: BRICS Cooperation in the Mineral Sector", *South African Journal of International Affairs* 32, 1-2 (2025): 215–241.
[110] Ibid.
[111] François Chimits and Grzegorz Stec*, The New European Commission and China + EU Trade with China and the US + EV Tariffs,* MERICS Briefs (18 October 2024), https://merics.org/de/merics-briefs/new-european-commission-and-china-eu-trade-china-and-us-ev-tariffs (accessed 7 July 2025).
[112] Victoria Waldersee and Christoph Steitz, "China's Rare Earth Export Curbs Hit the Auto Industry Worldwide", *Reuters Media*, 4 June 2025, https://www.reuters.com/business/autos-transportation/some-european-auto-supplier-plants-shut-down-after-chinas-rare-earth-curbs-2025-06-04/ (accessed 14 July 2025).

rivalry between China and the US, they are increasingly interpreted as instruments of leverage in China's negotiations, not only vis-à-vis Washington, but also with Brussels.[113] Consequently, at the G7 summit in Canada, Commission President von der Leyen adopted a significantly more critical tone, accusing China of following a "pattern of dominance, dependency, and blackmail" in its global trade conduct.

Despite mounting criticism and political frictions, a full decoupling from China in the minerals sector remains neither realistic nor strategically advisable.[114] China's dominance in critical mineral supply chains is deeply entrenched, and diversification efforts by the EU and its "like-minded" partners have been lacking. China continues to expand its engagement in the minerals sector across the Global South and focus on securing access to minerals. It is using outbound investment through both state-owned and private enterprises. This effort is underpinned by strong institutional backing from the China Export-Import Bank and the China Development Bank, which have collectively financed more than 70% of Belt and Road Initiative (BRI) projects—amounting to approximately $962 billion since 2013.[115] This comes amid analyses that China is now recalibrating BRI towards becoming a "small but beautiful" strategy. Although BRI investment levels have fluctuated, Chinese investments and new contracts in the raw materials sector reached a new peak in 2023. This approach is marked by close coordination between the government and investors, high-risk tolerance, and a long-term horizon, setting it apart from the EU's more cautious, recently renewed engagement in this space.

Yet, China's model is not without its critics. Many recipient countries have raised concerns about debt sustainability, non-transparent agreements, and limited local value creation. However, China has begun to respond strategically. Most recently, Beijing removed tariffs on African imports—framing itself as a committed trade partner and aiming to place greater emphasis on local value addition. For example, in Zimbabwe, it supports the "Mine-to-Energy Industrial Complex". In Chile, the Chinese automaker BYD has secured lithium supply agreements and plans to construct a cathode manufacturing plant in Antofagasta, although implementation has been delayed.

Given these developments, the EU must strike a careful balance between mitigating the immediate impacts of trade restrictions, actively pursuing de-risking, and protecting markets and strategic supply chains vis-à-vis China, while also identifying opportunities for pragmatic engagement.[116] This is especially relevant in the engagement with mineral-

[113] European Commission, *Statement by President Von Der Leyen at Session II - Working Lunch of the G7*, https://ec.europa.eu/commission/presscorner/detail/en/statement_25_1522 (accessed 14 July 2025); Antonia Zimmermann, "China's Got the World in a Rare Earth Choke Hold", *Politico*, 18 June 2025, https://www.politico.eu/article/china-rare-earth-materials-donald-trump-west-magnets-cars/ (accessed 14 July 2025).

[114] Inga Carry, Nadine Godehardt and Melanie Müller, *The Future of European-Chinese Raw Material Supply Chains: Three Scenarios for 2030 and Their Implications,* SWP Comment 2023/C 27 (Berlin: Stiftung Wissenschaft und Politik [SWP], May 2023), doi: 10.18449/2023C27.

[115] Jacob Gunter, Aya Adachi, Francesca Ghiretti and Gregor Sebastian, *China Inc Is Recalibrating Its Approach to Its Overseas Footprint as the BRI Turns 10 Years Old,* China Global Competition Tracker 3 (Berlin: Mercator Institute for China Studies, 27 September 2023), https://merics.org/en/tracker/china-inc-recalibrating-its-approach-its-overseas-footprint-bri-turns-10-years-old (accessed 25 June 2024). Christoph Nedopil, *China Belt and Road Initiative (BRI) Investment Report 2022*, (Shanghai: Green Finance & Development Center, FISF Fudan University, January 2023), https://greenfdc.org/wp-content/uploads/2023/02/Nedopil-2023_China-Belt-and-Road-Initiative-BRI-Investment-Report-2022.pdf (accessed 19 August 2024).

[116] Barbara Lippert and Stefan Mair (Eds.), "Neue Verhältnisse – schwierige Beziehungen" (see note 82); Alicia Vasselier and Abigaël García-Herrero, *Updating the EU Strategy on China: Co-Existence While Derisking Through Partnerships,* Policy Brief (Brussels: 7 July 2025), https://www.bruegel.org/policy-brief/updating-eu-strategy-china-co-existence-while-derisking-through-partnerships (accessed 7 July 2025).

rich countries in the Global South.[117] China remains a key partner for many of them, which often pursue non-alignment policies. At the same time, the global integration of Chinese firms—especially at the stages of extraction as well as smelting and refining—makes it difficult to exclude them from critical supply chain governance.

One relevant area is continued dialogue on standard-setting, both in multilateral fora and through international initiatives. This is particularly important as Chinese companies face ongoing criticism for human rights violations and environmental damage in the extractive sector.[118] Notably, China has shown some openness to addressing these concerns: Since 2015, the China Chamber of Commerce of Metals, Minerals & Chemicals Importers & Exporters (CCCMC) has collaborated with the Organisation for Economic Co-operation and Development (OECD) on developing the Chinese Due Diligence Guidelines for Responsible Mineral Supply Chains. There has also been punctual cooperation on sustainability and corporate responsibility in mineral supply chains with national agencies such as Germany's Federal Institute for Geosciences and Natural Resources (BGR) and GIZ.[119] Of course, such initiatives must be critically assessed and continuously monitored to ensure they serve the interests of both sides equally and are not used to advance adverse or one-sided agendas. Nevertheless, they provide a potentially valuable platform for technical cooperation and a basis for building trust between European and Chinese stakeholders. Additionally, in 2024, the Chinese Ministry of Finance announced plans to establish a unified national sustainability framework by 2030. Chinese firms have also joined the Initiative for Responsible Mining Assurance (IRMA).[120] Such engagement should extend to joint implementation in third countries. In Zambia, for instance, cooperation on tailings management or similar environmental practices could offer concrete and constructive entry points for collaboration.

Another promising area for engagement is infrastructure development. A narrow focus on geopolitical competition is often poorly received in partner countries, as it can undermine development goals or reduce the market feasibility of new projects by promoting exclusivity. In contrast, business actors on the ground typically adopt a more pragmatic approach than political decision-makers.[121] The growing agency of developing countries may also create new opportunities for more needs-driven engagement in addition to opening new pathways for more cooperative approaches. For example, in the DRC, pressure on Chinese investors has been rising. In 2023, Kinshasa began reviewing existing mining contracts in 2023 due to concerns over unfulfilled infrastructure commitments and has imposed a cobalt export ban in 2025 that primarily affects Chinese offtakes. These dynamics may push China towards greater openness to international cooperation to meet host countries' expectations. The Lobito Corridor between the DRC, Angola, and Zambia provides a concrete case. Chinese companies remain key producers in the region and are expected to use the transport corridor to make it viable—despite the

[117] Christian Géraud Neema, *The Tumultuous Path Toward EU-China-Africa Trilateral Cooperation on Critical Raw Materials in Africa*, 20 May 2025, https://afripoli.org/the-tumultuous-path-toward-eu-china-africa-trilateral-cooperation-on-critical-raw-materials-in-africa (accessed 7 July 2025).
[118] Business & Human Rights Resource Centre, *Unpacking Clean Energy: Human Rgihts Impacts of Chinese Overseas Investment in Transition Minerals* (July 2023), https://media.business-humanrights.org/media/documents/2023_China_TM_briefing.pdf (accessed 25 June 2025).
[119] Lutz Berners, *Sustainability Standards in the Chinese Rare Earth Supply Chain*, Final Report (Hannover: Bundesanstalt für Geowissenschaften und Rohstoffe, December 2024), https://doi.org/10.25928/0c8z-we31 (accessed 14 July 2025).
[120] Barbara Lippert and Stefan Mair (Eds.), "Neue Verhältnisse – schwierige Beziehungen" (see note 82).
[121] Poorva Karkare and Bruce Byiers, *The Lobito Corridor: Between European Geopolitics and African Agency*, Discussion Paper 386 (Brussels: ECDPM, April 2025), https://ecdpm.org/application/files/1217/4479/3017/The-Lobito-Corridor-Between-European-geopolitics-African-agency-ECDPM-Discussion-Paper-386-2025.pdf (accessed 25 June 2025).

project being geopolitically backed by the US and the EU. In all such cases, close cooperation with host governments is essential to assess what is technically and politically feasible.

In parallel to China's longstanding dominance, the Gulf States—particularly Saudi Arabia[122] and the UAE—are emerging as increasingly strategic actors in global mineral supply chains. Although they are traditional oil giants, they are positioning themselves in the minerals market through ambitious resource diplomacy and targeted investments. The return of the Trump administration has further elevated the role of the Gulf States as strategic partners and co-investors.[123] For the EU, cooperation with the Gulf offers opportunities for supply chain diversification, but it also entails challenges related to political dependencies and new vulnerabilities.

The Gulf States are currently pursuing a strategic expansion of their role in mineral value chains. This includes securing equity stakes in mining operations abroad, building up domestic processing capacities, and transforming themselves into global trade and logistics hubs. Similar to China's approach, these ambitions are state-driven and backed by sovereign wealth funds or by close ties between the state and major business actors.[124] With vast financial resources, these countries are acquiring or announcing significant shares along the supply chain. For example, in 2024, Saudi Arabia's investment vehicle Manara Minerals (a joint venture between the Public Investment Fund and Ma'aden, the semi-state-owned mining company) acquired a 10% stake in the metals division of Brazilian mining giant Vale, and it is pursuing further investments in countries such as Zambia, the DRC, and Chile.[125] On the UAE side, existing infrastructure strength is now being expanded into the critical minerals space. International Resources Holding, for instance, became majority owner of a Zambian copper mine in 2024 and also secured a deal for a tin mine in the DRC. However, critical questions persist regarding the firm's operational conduct, adherence to human rights and environmental standards, and the overall nature of their engagement with host countries.[126] Moreover, the UAE's involvement in the illicit trade of minerals—particularly gold sourced from Africa—undermines its credibility as a reliable and reputable trading partner.[127]

In this evolving context, the EU urgently needs a clear strategic posture and a more coherent engagement framework. Although the EU itself has not concluded direct bilateral agreements on raw materials, new developments are underway: The EU Council has taken

[122] Schulze and Schrolle, "Saudi Arabia Strives to Become Major Player in Mineral Supply Chains" (see note 80).

[123] Stephan Roll and Antonia Thies, *Saudi Arabia's 'Vision 2030' and Trump's Second Term*, 13.01.2025, https://www.swp-berlin.org/publikation/saudi-arabias-vision-2030-and-trumps-second-term (accessed 14 July 2025).

[124] Gracelin Baskaran, *Saudi Arabia Has a Strategic Advantage in Sourcing Critical Minerals from Africa,* Commentary (5 December 2023), https://www.csis.org/analysis/saudi-arabia-has-strategic-advantage-sourcing-critical-minerals-africa (accessed 7 July 2025); Lee Bailey, Laury Haytayan and Thomas Scurfield, *As Saudi Arabia and the UAE Expand Foreign Mining Interests, How Can Producing Countries Prepare?*, 17 September 2024, https://resourcegovernance.org/articles/saudi-arabia-and-uae-expand-foreign-mining-interests-how-can-producing-countries-prepare (accessed 30 June 2025).

[125] Schulze and Schrolle, "Saudi Arabia Strives to Become Major Player in Mineral Supply Chains" (see note 80).

[126] Ibid.

[127] Sasha Lezhnev and Megha Swamy, *Understanding Money Laundering Risks in the Conflict Gold Trade: From East and Central Africa to Dubai and Onward* (Washington, D.C.: The Sentry, November 2020), https://thesentry.org/wp-content/uploads/2020/11/ConflictGoldAdvisory-TheSentry-Nov2020.pdf (accessed 25 June 2025); Global Initiative Against Transnational Organized Crime, *Trade and Transit: Dubai's Role in Illicit Environmental Supply Chains* (December 2022), https://globalinitiative.net/wp-content/uploads/2022/12/Trade-and-transit-Dubais-role-in-illicit-environmental-supply-chains-GI-TOC-2022.pdf (accessed 25 June 2025).

an initial step by deciding to launch negotiations for bilateral strategic partnership agreements with the six countries of the Gulf Cooperation Council (GCC), building upon the momentum of the October 2024 EU-GCC summit. However, although economic considerations support closer cooperation, the EU remains internally divided over deeper bilateral cooperation—particularly in the raw materials sector.[128] Individual member states are moving ahead, such France and Austria, which have signed MoUs with Saudi Arabia in the field of critical raw materials.

Failing to engage with the Gulf's strategic initiatives is neither realistic nor advisable in the context of the EU's efforts to diversify its mineral supply chains. However, establishing full-fledged CRM strategic partnerships with Saudi Arabia or UAE, particularly those involving public co-investment through entities such as the EIB, would be ill-advised. This is due to the high risks posed by heavy state involvement and the potential for political interference, as well as insufficient standards in terms of transparency and accountability.[129]

Despite these risks, the Gulf region is gaining relevance in global supply chain rerouting. Its financial capacity and expanding role as a minerals hub make selective engagement increasingly necessary. The EU should therefore pursue an interest-driven strategy that emphasises transparency and responsible sourcing, particularly if it aims to become a larger off-taker of critical minerals. This means that the EU should continue to engage in multilateral platforms such as the IGF and encourage Gulf-based actors to align with private-sector certification schemes such as the Copper Mark, which offer pragmatic entry points for collaboration. Inclusion in engagement at the country level, such as with China (see above), could further support improved sector governance.

Recommendations

- **Build a G7-based "coalition of the willing" that is focused on strategic coordination and project implementation.** The EU should reinforce practical collaboration within existing frameworks such as the G7 and the Minerals Security Partnership, prioritising like-minded and implementation-oriented partners such as Canada and Australia. Rather than launching new formats, the focus should be on outcome-driven cooperation and expanded engagement with resource-rich countries in other regions. Mobilising private capital requires streamlined permitting and de-risked investment conditions in exploration and mining, anchored in high environmental and social standards. Equally important is supporting vertical integration by facilitating refining and processing capacities in partner countries in the Global South to foster local value creation and supply chain resilience.
- **Pursue strategic, credible, and partner-oriented engagement with the Global South.** The EU should adopt a realistic engagement approach that reflects geopolitical complexity and avoids conflict-driven narratives. Political commitments must be backed by real implementation capacity, financing, and high-impact projects with strong private-sector participation. Rather than multiplying MoUs, existing commitments should be prioritised, with raw materials being integrated into broader areas of cooperation such as energy and infrastructure. Regional collaboration should be fostered—particularly through

[128] Background conversation with EU public officials in September 2024 and February 2025.
[129] Schulze, "Security of Supply in Times of Geo-economic Fragmentation" (see note 74).

platforms such as AfCFTA—and development cooperation should support good governance and institutional capacity. The EU must also pragmatically accommodate national development strategies, especially where they target local value addition and industrialisation.

- **Develop clear and differentiated EU strategies for cooperation with complex partners, grounded in a consistent and coordinated foreign policy framework.** This includes promoting further diversification away from China, backed by the private sector, while avoiding an outright decoupling that would limit opportunities for pragmatic cooperation. Although reducing dependencies on China remains essential—particularly through private-sector alignment—pragmatic engagement should continue where interests align. At the same time, rising dependency risks from new influential players, such as the Gulf States, should be assessed and addressed. The EU should actively promote transparency and ESG dialogues with these actors, bilaterally and multilaterally, to prevent a global "race to the bottom" in third-country contexts.
- **Engage multilaterally to reduce the risk of bloc-based confrontation and competition in the raw materials sector.** The EU should promote its interests in responsible sourcing primarily through multilateral platforms and fostering broader legitimacy, cooperation—particularly with the Global South— and stronger engagement in arising opportunities such as the G20, the UN, and other forums for cooperation.

5 Conclusion and Policy Recommendations

The EU stands at a strategic crossroads. In this time of intensified geopolitical rivalry, economic coercion, and fragmenting global governance, Europe's dependency on critical raw materials—particularly those from China—has become a central vulnerability. These dependencies are no longer merely economic in nature; they carry systemic risks for Europe:

- They expose the EU to political pressure and coercion, as raw materials become increasingly weaponised in geopolitical competition.
- They undermine the industrial resilience and technological competitiveness of European companies.
- They jeopardise the implementation of the twin transition and defence agenda, which depends on uninterrupted access to critical inputs.
- They threaten the credibility and effectiveness of European defence efforts, given that capabilities cannot be scaled without resilient and secure supply chains.

With the adoption of the Critical Raw Materials Act in May 2024, the EU has made an important step towards greater strategic autonomy. The CRMA provides a necessary policy framework for boosting domestic capacity and diversifying external supply. Raw material supply chains are complex, and building resilient, sustainable value chains is a long-term, high-risk endeavour—one that requires more than policy guidance. It calls for political resolve, substantial financing, and international coordination. Raw materials partnerships are key to achieving the goals set out in the CRMA; however, many of them have yet to gain sufficient industrial traction and reliable financing to really make use of their potential and value. Strengthening external coordination across instruments and developing a clear geopolitical strategy that takes into account not only the EU's but also partner countries' interests is essential. Otherwise, there is a risk that the EU could lose ground to more assertive players such as China or the Gulf States. Crucially, none of this can succeed without active and sustained private-sector involvement. Yet, too many EU lead firms still operate under outdated assumptions about supply stability and predictable market conditions, despite the fact that today's geopolitical and supply chain dynamics demand greater risk affinity and long-term strategic commitment.

To build a truly resilient system, the EU must link its internal industrial ambitions with a coherent external engagement strategy. Domestic efforts alone will not be sufficient in an environment of intensified competition and growing resource nationalism. This is particularly true for the raw materials sector, which is shaped by structural characteristics that limit the effectiveness of short-term or purely national responses. Long lead times, the spatial fixation of resource deposits, and the capital-intensive nature of extraction and processing make it difficult to respond quickly to supply disruptions. Recent export restrictions have demonstrated just how challenging it is to address such shocks through emergency measures alone.

Therefore, it is of upmost importance for the EU not to wait until the next crisis emerges. Instead, it must act now to advance medium- and long-term strategies that strengthen its strategic position. Legislative texts and partnerships now need a strategic follow-through that is proactive, not reactive, and anchored in geopolitical awareness and operational delivery. Without a consistent and credible external raw materials strategy, the EU risks losing the trust of partner countries and limiting its ability to shape future supply chains. The following section summarises the most important policy recommendations for developing a forward-looking approach that aligns industrial and financial tools, supports co-investment in partner-led value creation, and invests in multilateral formats based on shared standards.

1. Understanding the Priorities of Partner Countries: Local Value Addition and Regional Strategies

The EU should anchor its raw materials partnerships in a clearly defined and jointly developed industrial perspective. To be credible, the EU must approach terms such as "local value addition" and "win-win partnerships" with care. If they are used vaguely or insufficiently backed by practical measures, they risk creating unrealistic expectations and damaging trust. Instead, what is needed is long-term, transparent cooperation that connects the raw materials agenda with wider development and industrial goals across sectors and regions.

- **Mutual benefit as a foundation**: Effective cooperation requires a better understanding of partner countries' industrial strategies and capacities. Raw materials partnerships must serve not only the EU's security of supply, but also partner countries' goals for industrialisation and diversification. This includes closer coordination between EU institutions, member states, and the private sector.
- **"Beyond extraction" enabling production linkages**: Many resource-rich countries aim to use the rising demand for critical minerals to promote local value addition. Linkages between mining and up-/downstream sectors can help move beyond "extractivism". However, their viability depends on geological, technological, and macroeconomic conditions. To avoid reinforcing mono-sectoral dependence, equal attention should be given to horizontal linkages across other sectors.
- **Strategic use of existing instruments**: The EU should better align its various cooperation tools—raw materials partnerships, various TEIs, JETPs, and CTIPs—and use them to identify cross-sectoral synergies, for example, with the energy, transport, and defence sectors.
- **Support for regional strategies:** Regional approaches are becoming increasingly relevant in raw materials governance. Although coordination capacities vary, regional cooperation can support shared infrastructure, industrial scaling, and joint standard-setting. In cases where partner countries express interest, the EU should complement bilateral partnerships with targeted bi-regional engagement. This can strengthen strategic alignment and help counter rising competition and protectionist dynamics among resource-rich states.

2. EU's External Raw Materials Strategy: Strengthening and Streamlining EU Efforts

The EU must consolidate its raw materials ambitions into a coherent external strategy. This involves not just institutional coordination, but also embedding raw materials

cooperation into broader investment and trade, climate, and development policy frameworks aligned with the EU's long-term strategic interests.

- **Position raw materials policy as a strategic branch of EU external action**: A credible raw materials agenda must be embedded in broader foreign policy thinking. This requires visible leadership from the European Commission and coordination across DG GROW, DG TRADE, EEAS, DG INTPA, and the CRM Board. Only a politically empowered and well-coordinated external pillar can establish trust with partners and translate MoUs into action.
- **Avoid protectionist reflexes**: A successful raw materials strategy needs to establish a careful balance between strengthening domestic extraction and processing within the EU and genuine support for local value creation in partner countries. Building up strategic capacities at home is important, but it should not undermine cooperation with countries that are themselves trying to industrialise. If the EU wants to be seen as a credible and long-term partner, it must invest not only in its own supply security, but also in partnerships that reflect the economic priorities and development strategies of producing countries.
- **Streamline and align external engagement**: The EU has taken steps towards a more unified external presence. Yet, fragmented bilateral activities and funding opportunities still weaken impact. A genuine Team Europe approach—anchored in shared objectives, joint financing, and common messaging—is essential to enhance impact on the ground.
- **Private sector and finance:** Europe's raw materials strategy remains incomplete without visible industrial support and predictable funding. This undermines both its credibility in partner regions and its ability to strengthen supply chain resilience.
    - **Strengthen the industrial backbone of partnerships**: Political frameworks alone are not sufficient. Industrial participation in upstream and midstream activities abroad remains limited. The EU should define clearer incentives (e.g. de-risking tools, long-term offtake structures) and explore regulatory instruments (e.g. diversification and stockpiling requirements) to align private-sector engagement with strategic supply goals.
    - **Promote shared responsibility:** The EU must send a stronger signal to companies and business associations to take de-risking seriously. A business-as-usual approach, shaped by assumptions about past market stability, is no longer viable. Although the EU and member states can create the right framework conditions for more proactive engagement, companies reliant on the supply of CRMs must equally step up with long-term action to ensure the resilience of their supply chains.
    - **Strengthen the financing pillar:** Financing remains a weak link in the EU's external raw materials strategy. Better integration and streamlining of member states' financial tools is needed. A financially robust CRM fund at the EU level would be the most effective measure to unify scattered instruments, facilitate access to finance, accelerate delivery, and signal a commitment of resources to key partner countries and CRM value chains.

3. Navigating a Fragmented International Order

In a contested global landscape, the EU must position itself as an autonomous yet reliable and partnership-oriented actor. This includes deepening cooperation with like-minded countries, continuing to invest in multilateral frameworks, and advancing trilateral formats. Engaging complex partners through principled yet pragmatic frameworks will be key to building sustainable and resilient supply chains.

- **Send clear signals to partners and competitors:** The EU needs to set and uphold its principles—defending the rules-based (economic) order and its commitment to sustainability standards—while at the same time asserting what makes it an attractive and reliable partner in its own right. Rather than defining itself primarily in opposition to others, the EU should focus on highlighting its value proposition and distinct approach to partnerships.
- **Invest in multilateral formats**: As traditional alliances shift, the EU should focus on building a coalition of like-minded partners who share its geopolitical goals and commitment to a rules-based order and strong sustainability standards. Instead of trying to do everything alone, closer coordination with countries such as Canada and Australia—both experienced in the mining sector—could help create a more targeted approach, pool financing, and align ESG standards. At the same time, forums and institutions such as the G20, the IEA, and the OECD provide important spaces for setting standards, increasing transparency, and coordinating on critical minerals. The EU would do well to engage more actively in these groups to help shape global norms and prevent further fragmentation.
- **Engaging complex partners:** A pragmatic, standards-driven approach is essential to navigate the balance between geopolitical realities and the EU's goals for diversification and resilience. When it comes to China, the focus should be on de-risking rather than full decoupling, avoiding a downward spiral in standard-setting by engaging selectively and strategically. Similarly, with the Gulf States and other emerging investment partners in the minerals sector, the EU should actively shape cooperation by providing clear guidance to the private sector that is grounded in principles such as transparency and due diligence to ensure these relationships align with shared values and long-term objectives.

Dr Melanie Müller is Senior Associate and Deputy Head of the Africa and Middle East Research Division at SWP. She also serves as Co-Head of the Research Network Sustainable Global Supply Chains.

Inga Carry is an Associate at SWP and a researcher in the project Research Network Sustainable Global Supply Chains.

Meike Schulze is an Associate at SWP and researcher in the project International Raw Materials Cooperation.

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SWP
Stiftung Wissenschaft und Politik
German Institute for International and Security Affairs

Ludwigkirchplatz 3–4
10719 Berlin
Telephone +49 30 880 07-0
Fax +49 30 880 07-100
www.swp-berlin.org
swp@swp-berlin.org

Working Paper

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Appendix

EU and German Raw Material Partnerships and Cooperation Projects

EU Raw Material Partnerships

Lead: DG GROW and DG INTPA
Countries:
Australia, Uzbekistan, Norway, Rwanda, Greenland, DR Congo, Zambia, Chile, Argentina, Namibia, Kazakhstan, Ukraine, Canada, Serbia

German Raw Material Partnerships

Lead: BMWK
Partnerships:
Kazakhstan, Mongolia, Peru, Chile

Germany International Raw Material Projects

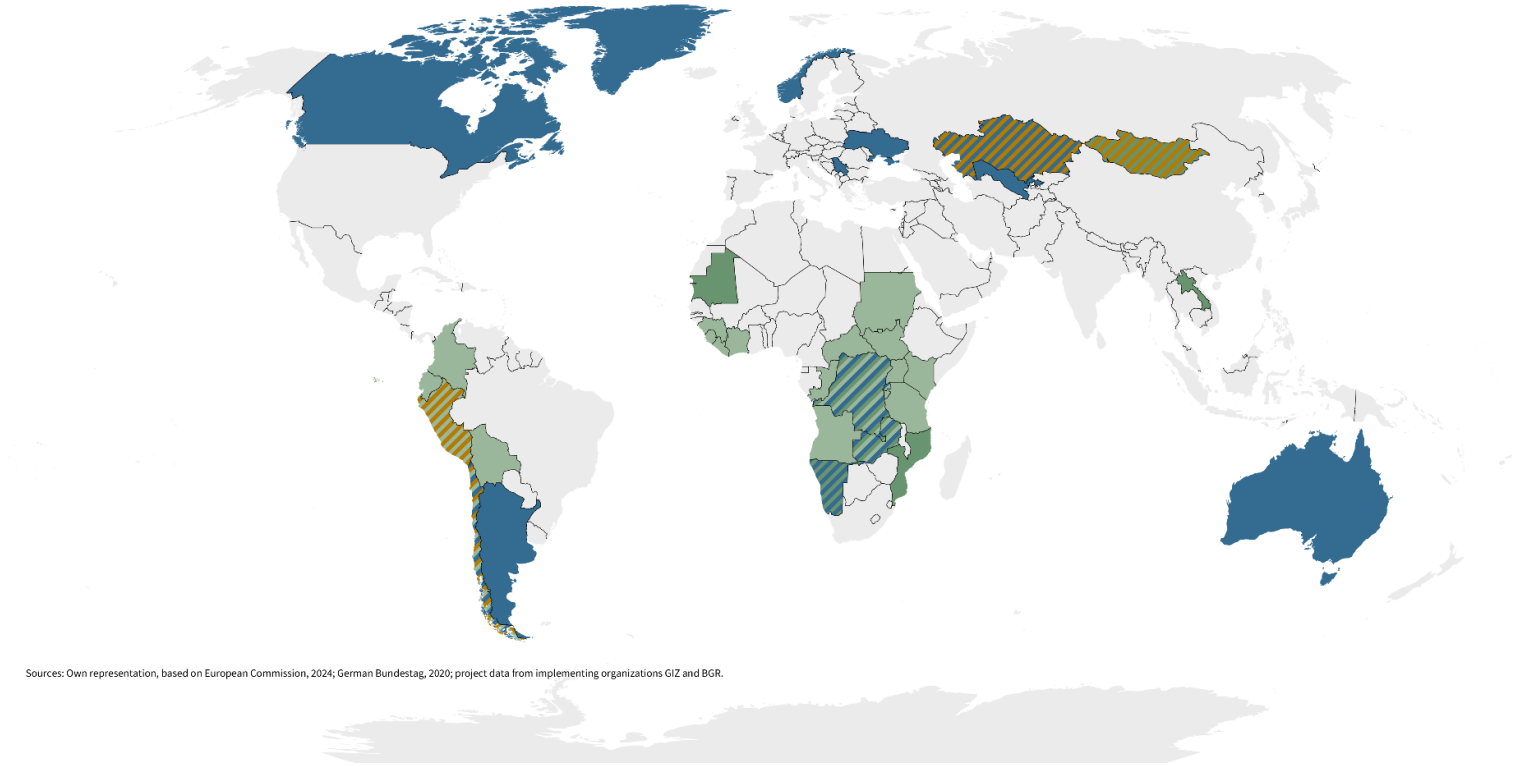
Commissioned by: BMZ, BMU
Additional projects include, e.g., CONNEX, AfricaMaval

Bilateral Projects
Laos, Mongolia, Mauritania, DR Congo, Namibia, Zambia, Mozambique

Regional Projects
West Africa (Côte d'Ivoire, Guinea, Liberia, Sierra Leone), Andean Region (Chile, Bolivia, Colombia, Ecuador, Peru), African Great Lakes Region (Angola, Burundi, Kenya, DR Congo, Republic of Congo, Rwanda, Zambia, Sudan, South Sudan, Tanzania, Uganda, Central African Republic)

Status: May 2024
This map is for illustrative purposes. It does not reflect any position on international legal claims. For optimal readability, a colored representation is recommended.

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Sources: Own representation, based on European Commission, 2024; German Bundestag, 2020; project data from implementing organizations GIZ and BGR.